When Volatility Masquerades as Fragility: DeFi Liquidations, Quantile Local Projections, and the Tails of Ethereum Returns

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July 6, 2026

Abstract

Decentralized-finance (DeFi) liquidations record forced deleveraging transaction by transaction in public data (a granularity unavailable in centralized venues, where the full mechanism is largely opaque). The leverage-cycle literature predicts that such forced selling should amplify downside tail risk. We test this prediction for Ethereum (ETH) using quantile local projections on 40 582 hourly observations (2021–2025). A naive specification appears to confirm it: the response of the first return percentile to a liquidation shock is roughly 18.8× the median response on impact, deepens with the horizon, and survives conventional robustness checks with Bonferroni correction in full. A battery of diagnostics overturns this reading. A symmetric sign-flip placebo reproduces the entire left–right tail gap at every horizon; the response is equally strong in the *upper* tail; the same signature appears when Bitcoin is substituted as the outcome (at 0.60–0.70× the magnitude); and dual permutation nulls show the long-horizon coefficient is genuine rather than a mechanical artifact of overlapping cumulative returns. Net of volatility, an equivalence test *bounds* the one-percent-tail down-minus-up asymmetry below economically negligible thresholds under all three calibrations of the shock span, including the strictest; only a small standardized residual is detectable, and it too is negligible (borderline under the strictest calibration). What survives is a *symmetric* volatility channel: liquidations carry out-of-sample predictive content for both moderate return tails at short horizons, incremental to a nested quantile benchmark though not a substitute for a dedicated volatility model. The exercise yields a

*Replication code and data available at https://github.com/Paul-Engerran/DeFi-Endogenous-Fragility. All figures, tables, and in-text numbers are generated from canonical artefacts by a one-command pipeline (`make paper`).

portable methodological caution. In quantile local projections on overlapping cumulative returns, symmetric heteroskedasticity masquerades as downside fragility.

Keywords: DeFi, liquidations, quantile local projections, tail risk, leverage cycle, equivalence testing.
JEL: G12, G14, G23, C22, C58.

1 Introduction

Hourly Ethereum (ETH) returns carry a marked downside tail asymmetry. Over 2021–2025 the skewness is -0.63 with excess kurtosis of 17.6, the worst hour (-15.5%) is far larger than the best ($+9.2\%$), and hours below -5% outnumber hours above $+5\%$ two to one (50 versus 25).[1] For equities, the textbook reading of such asymmetry runs through the leverage effect. In crypto, that channel is often found to be weak, absent, or inverted (Baur and Dimpfl, 2018; Brini and Lenz, 2022; Aharon et al., 2023), and a commonly invoked alternative mechanism is the liquidation cascade of the leverage cycle (Geanakoplos, 2010; Brunnermeier and Pedersen, 2009), in which falling prices force collateralized positions to unwind, and forced selling begets further falls. The mechanism is plausible and widely invoked, both in empirical studies of DeFi leverage and liquidation risk (Perez et al., 2021) and in official-sector financial-stability assessments (Sasi-Brodesky and Nassr, 2023; Financial Stability Board, 2023). In centralized venues the full mechanism is largely opaque.

Decentralized finance changes the observability. DeFi lending protocols liquidate undercollateralized positions mechanically, on-chain, transaction by transaction. The shadow quantity at the heart of equilibrium leverage models, the tightening of collateral constraints, is recorded in public data. We assemble an hourly panel of ETH-collateralized DeFi liquidations (2.52 billion USD across 2021–2025) and ask whether this observable forced deleveraging amplifies the downside tail of ETH returns, as the leverage-cycle prior predicts. We are explicit about the design. The measure is aggregated to the hour, covers DeFi only (centralized-exchange liquidations are out of scope), and at short horizons the liquidation–return link is mechanically bidirectional: 85.5% of the worst-1% return hours contain contemporaneous liquidations, against 41.6% of the best-1% hours and 27.0% unconditionally. Two coverage gaps further bound the claim: MakerDAO, historically among the largest liquidation venues,

[1] The asymmetry is a *far-extreme* property: counts beyond absolute $\pm5\%/\pm7\%$ thresholds. The 1% and 5% quantiles themselves are nearly symmetric (ratio 1.05 at the 1% level), a distinction that matters below. The sign of measured crypto skewness differs across return definitions and samples: in simple returns it is positive across major cryptocurrencies (Liu and Tsyvinski, 2021), while the hourly log returns studied here skew negative. Fat tails are the expected stylized fact (Cont, 2001; Osterrieder and Lorenz, 2017).

is de facto absent from the panel, and the sample predates the liquid-restaking-token collateral generation; we assess the direction of both in Section 9. The register of the paper is therefore descriptive and predictive, never causal.

Our empirical vehicle is the quantile local projection (QLP) of Han et al. (2024), the natural transposition of the Growth-at-Risk logic of Adrian et al. (2019), in which tails can respond where the mean is silent, to high-frequency crypto returns. The naive QLP delivers what the prior predicts, and what this nascent literature could plausibly report before recognizing the artifact. On impact, the response of the first return percentile to a liquidation shock is 18.8× the (near-zero) median response; the coefficient deepens monotonically to -0.241 at twelve hours; the left tail appears to respond more than the right; and the pattern survives a robustness battery with Bonferroni adjustments. It looks like endogenous fragility. It is not. Showing why, with a diagnostic battery that travels beyond crypto, is the contribution of this paper.

The diagnostics dismantle the downside-amplification reading and replace it with something more modest and, we argue, more useful. First, a *symmetric* placebo, a sign-flip world that preserves the exact empirical volatility path and the full magnitude distribution of returns but imposes zero skew by construction (Wu, 1986; Liu, 1988; Davidson and Flachaire, 2008), reproduces the entire left–right tail gap at every horizon, including the deepening profile. The apparent downside specificity is what symmetric heteroskedasticity looks like through the lens of extreme quantiles, the mechanism identified in macroeconomic data by Carriero et al. (2024) and, for scale versus shape in quantile regression, by Koenker and Bassett (1982) and Machado and Silva (2000). Second, the clean per-period counterpart of the naive regression, tail-exceedance indicators in the spirit of Engle and Manganelli (2004), shows the shock loads *equally on both tails* ($\Delta = \beta^{\text{down}} - \beta^{\text{up}} = 0.00053$ at the 1% level on impact, $p = .110$), while the overlapping cumulative variant manufactures a significant asymmetry at twelve hours: the artifact made visible in a single comparison (Hodrick, 1992; Boudoukh et al., 2008; Neuberger and Payne, 2021; Farago and Hjalmarsson, 2023). Third, re-running the full QLP with Bitcoin as the outcome reproduces the ETH signature at 0.60 of its magnitude at twelve hours, with the same deepening: the pattern is generic crypto stress, not an ETH-specific DeFi channel. Fourth, an equivalence test in the spirit of Altman and Bland (1995) and Lakens (2017) converts the null into a bound. Net of volatility, we can bound two objects. The one-percent-tail exceedance asymmetry (down minus up) is below our pre-specified smallest effect size of interest under all three calibrations of the shock span, including the strictest; a small standardized residual is statistically detectable but economically negligible, clearing two of the three calibrations and borderline under the strictest. Jointly they exclude downside-specific amplification larger than 0.00094 per unit of the log-liquidation shock. This bound is a one-percent-tail statement; it does not constrain the far-extreme (±5–7%) asymmetry documented above, which an hourly panel cannot resolve.

What survives is worth having. Liquidations predict ETH tail risk in both tails, symmetrically. Dual permutation nulls (a circular shift of the shock and an innovation reshuffle) confirm that the long-horizon coefficient reflects signal rather than overlap noise, with no signed bias. And the predictive content holds out of sample. In a rolling-origin pinball-loss evaluation (Diebold and Mariano, 1995), adding liquidation information to an otherwise identical quantile model improves tail forecasts at both moderate tails at short horizons (by 1.0% and 0.9% at one hour, $p < 0.01$), though it does not outperform a dedicated GARCH(1,1) scale model; the claim is incremental information, not dominance. The volatility channel itself confirms and extends the transaction-level evidence of Sasi-Brodesky and Nassr (2023) and Lehar and Parlour (2022) to the conditional tails.

The contribution is a portable *methodological caution*, and it materializes in a new empirical regularity. In quantile local projections on overlapping cumulative returns, a symmetric volatility response masquerades as downside fragility. The conjunction of extreme quantiles, cumulative overlap, and heteroskedastic returns is where the artifact arises, and the diagnostic battery we provide is built to detect it. Carriero et al. (2024) documented the mechanism for macroeconomic tail risk; we extend it to asset returns, where the policy narrative it can mislead ("DeFi liquidations amplify crashes") is live.

We mean the battery less as a one-off diagnosis of the DeFi data than as a portable test-bench: a symmetric placebo for shape, dual permutation nulls for level, a per-period-versus-cumulative exceedance comparison, standardized skew and equivalence bounds, a cross-asset outcome placebo, and an out-of-sample evaluation. Any quantile-local-projection study on overlapping cumulative returns can rerun it to separate scale from shape. That the apparatus is worth porting is itself a finding. It survives the adversarial repairs we put to it, and its positive control confirms the equivalence test recovers an asymmetry of actionable size while staying silent below it, so a bounded null it returns is power-confirmed, not power-absent.

The regularity the caution leaves standing is the positive finding: observable forced deleveraging is a *symmetric* out-of-sample predictor of ETH tail risk (incremental pinball skill of 1.0% and 0.9% at one hour, $p < 0.01$), to our knowledge among the first quantile-resolution evidence on on-chain forced deleveraging. The null it delivers is itself informative because it is *bounded*: net of volatility we exclude any downside-specific amplification larger than 0.00094 per unit of shock at the one-percent tail (the exceedance asymmetry clears all three calibrations, including the strictest; the small standardized residual clears two and is borderline under the strictest), a falsifiable statement rather than a bare failure to reject. The whole is made reproducible by an hourly on-chain liquidation panel (2021–2025) and a one-command replication package.

Why does observable forced deleveraging fail to amplify the downside? Scale and clearing. Median daily DeFi liquidations are 2×10^{-5} of one percent of ETH spot-plus-perpetual turnover; the worst liquidation day of the sample reaches 0.67%; and Moallemi and Patange (2024) document that over 70% of liquidations occur without any downward price jump, consistent with the near-instant clearing documented by Perez et al. (2021) and with the small, transient pool-level price impacts in Tian and Zhu (2025). The null is therefore explicitly *size-contingent*. It characterizes the current scale of observable DeFi deleveraging, not a structural impossibility, and would bear re-examination if DeFi's share of crypto leverage grows materially (Financial Stability Board, 2023). Likewise, nothing here speaks to liquidation cascades on centralized derivatives venues, which our design cannot observe. The asymmetry of ETH returns is real; our evidence says only that observable DeFi liquidations show no downside-specific amplification of it net of volatility.

The paper proceeds as follows. Section 2 describes the data and the construction of the liquidation measure. Section 3 establishes the stylized facts. Section 4 lays out the locked specification and inference protocol. Section 5 presents the naive result. Section 6 deconstructs it. Section 7 presents what survives, including the equivalence bounds and the out-of-sample evaluation. Section 8 discusses mechanisms, Section 9 scope and limitations, and Section 10 concludes.

2 Data and Institutional Background

2.1 DeFi liquidations as an observable deleveraging signal

Decentralized lending protocols (Aave, Compound, Spark, Morpho, and related systems) enforce collateralization rules through smart contracts. When a borrower's collateral value relative to outstanding debt crosses a protocol-defined threshold, any third party may repay part of the debt and seize an equivalent amount of collateral at a discount. The event is mechanical. It is triggered by price movement, not by borrower decision, and under the fixed-spread designs of these protocols it settles on-chain within a single block (Qin et al., 2021). Each liquidation is therefore an observable instance of forced deleveraging in the sense of Geanakoplos (2010) and Brunnermeier and Pedersen (2009): the object that equilibrium credit models treat as a shadow quantity, the tightening of collateral constraints, is here recorded transaction by transaction.

Two caveats qualify this observability at the outset. First, liquidator behavior after seizure is not observed. Whether seized ETH is resold immediately, consistent with the liquidator's incentive to realize the bonus, or held instead, is a behavioral assumption, and the seized-collateral measure is retained as a robustness alternative to the debt-repaid measure used below. Second, the hourly aggregation imposed by the data source does not resolve within-hour sequencing; the paper makes no claim about the order of events inside a given hour.

2.2 Panel and scope

The dataset is an hourly UTC panel from 2021-03-15 00:00 to 2025-12-01 00:00 (exclusive), a raw grid of 41 328 hours with timestamps marking the start of each bucket. Initializing rolling statistics consumes a 746-hour warmup, leaving an estimation panel of 40 582 hours. All sources are public: Bybit (perpetual prices, open interest, funding), CCData spot aggregates (ETH, BTC, and the placebo assets XRP and DOGE), Coinbase and Binance feeds used for cross-venue audits, and Dune Analytics for on-chain liquidations.

Two scope choices are deliberate. The hourly frequency is the identification compromise. It is fine enough to separate a one-hour-ahead lead from within-bucket simultaneity, and coarse enough that buckets carry non-trivial liquidation mass. Daily aggregation would drown any cascade dynamics, while tick-level sequencing is unavailable from the liquidation source. The on-chain perimeter is an *observability* constraint, not a convenience. DeFi is the only segment where forced deleveraging is recorded transaction by transaction. Liquidations on centralized derivatives venues, where a reflexive spiral, if any, plausibly lives, cannot be observed at this grain; they are neither tested nor refuted here (Section 9). The window starts when joint coverage of Bybit derivatives data and the Dune lending tables becomes adequate; the sample is post–DeFi-Summer and excludes the March 2020 episode.

2.3 The liquidation series

Liquidations are extracted from Dune Analytics (query `6912877`, versioned in the replication package) over the open-source Spellbook tables `lending.borrow` and `lending.supply`, covering all EVM chains and lending protocols indexed at extraction time (Aave v1–v3, Compound v2–v3, Spark, and Morpho account for the bulk of volume). The query was executed once, and the deposited hourly panel, not a live re-query, is the reproducibility anchor; the source-to-panel path is therefore a snapshot and is not end-to-end re-executable. The primary measure is *debt forcibly repaid*, in USD, aggregated by hour. This is the leg that measures the deleveraging flow itself, how much leverage was forcibly purged, rather than collateral seized, which mixes in the liquidator's discount. Perez et al. (2021) show that ninety-five percent of liquidable collateral is liquidated within sixteen blocks, a few minutes at most, which legitimates hourly aggregation: a liquidation is an event within the hour, not a process spanning hours.

Four filters make the measure conservative and economically interpretable. First, only liquidation-type events enter. Second, collateral is restricted to an ETH-like whitelist (WETH, ETH, stETH, wstETH, rETH, cbETH, sfrxETH, ETHx), with two caveats: liquid-staking derivatives are included, so the collateral set is ETH-like in the broad sense; and the list predates the liquid-restaking generation (weETH, ezETH, rsETH, osETH, and kin), which became material lending collateral only in 2024–2025 and is therefore absent. Because the collateral leg gates the join, this omission drops the affected liquidations entirely, an under-count bounded below. Third, a dust filter (absolute debt below \$5) removes a cluster of roughly 75,000 economically immaterial bot-driven micro-events in September 2025. Fourth, exploit windows are excluded (UwuLend from June 2024; Euler v1, March–April 2023), since exploit-driven liquidations are endogenous to a hack, not to the market cycle. Event tables are pre-aggregated by transaction before joining the debt and collateral legs, eliminating a Cartesian-product inflation that affects bundled liquidation transactions. Full query logic and exclusions are documented in the data appendix.

Three structural omissions bound the interpretation. First, the source aggregates to the hour: within-hour sequencing of prices and liquidations is lost, so at short horizons the liquidation–return relation is mechanically bidirectional. This is the basis for the paper's non-causal register (Section 3). Second, MakerDAO, historically the largest on-chain liquidator, sits outside the Spellbook lending schema (a CDP architecture) and is de facto absent, so the measure understates total on-chain deleveraging, most severely in 2021. Third, no centralized-exchange liquidations enter: the shock is DeFi-only.

A direct check bounds what the MakerDAO omission can distort. We re-fit the locked specification on a panel inflated by a missing-Maker mass (the displacement figures below are the maximum over the defensible allocation share $m \leq 0.25$; extending the sweep to a punitive $m = 0.5$ changes neither sign nor significance), allocated three economically credible ways: proportional to active liquidations, concentrated in stress-adjacent hours, and weighted toward high-leverage hours. The shock coefficient's sign stays intact under every allocation, and the impact level moves by at most 24%. The level magnitude at *medium* horizons is coverage-sensitive. Under a leverage-weighted infill, the largest-displacement allocation, since Maker liquidations cluster in high-leverage hours that are mechanically correlated with the down tail, the twelve-hour deepening *shrinks* by as much as 65%, so part of the apparent deepening is itself a data-coverage artifact, compounding the overlap artifact of Section 6.4, rather than a property of the cycle. The object the paper's conclusion rests on is more robust: the per-period exceedance asymmetry Δ at the 1% level (Section 7) stays within

0.48 of the smallest effect size of interest under every allocation, and moves *toward* zero under the realistic ones, so better Maker coverage would, if anything, sharpen the null. The full re-fit, the three allocations, and the per-horizon displacements are reported in Appendix C.

A second coverage gap operates the same way. The liquid-restaking tokens that came to dominate leveraged ETH lending in 2024–2025 fall outside the collateral whitelist, and because the collateral leg gates the join, liquidations seizing only those tokens drop from the measure entirely. The resulting under-count is one-sided and concentrated in the final third of the sample, the same window in which the leverage-cycle interaction is identified. As with the Maker omission, the direction is favorable: under-coverage of the shock is attenuating measurement error in the regressor, biasing the interaction and tail-asymmetry estimates toward the null the paper already reports. One caveat resists this logic, and we flag rather than dismiss it: restaking-token de-pegs (ezETH in April 2024, for instance) are not generic deleveraging but sharp, clustered, downside events, so the missing mass may be left-tail-selective rather than symmetric. On present evidence this does not overturn the deconstruction, which rests on properties of the return distribution and the estimator established independently of the shock's exact magnitude (Sections 6 and 7); closing it fully requires re-estimation on a collateral universe expanded to the restaking generation, which we leave to future revision.

Total qualifying liquidations amount to \$2.52 billion over the sample. The flow is extremely sparse and right-skewed: 27.0% of estimation-panel hours contain any liquidation; the mean daily flow is \$1.46 million against a median of \$9.3 thousand (both computed over all calendar days, not only active days); and the largest single day (\$307 million, 2024-08-05, the yen carry-trade unwind) sits more than three orders of magnitude above the median day (Figure 1). These features motivate the logarithmic shock transformation below. The external order of magnitude is consistent with the \$2.49 billion over 42,324 liquidations that Lehar and Parlour (2022) report through mid-2022.

2.4 Returns and market state

The dependent variable is the hourly log return of the Bybit ETHUSDT perpetual, $r_t = 100 \cdot \ln(P_t/P_{t-1})$. We select Bybit on a cross-venue diagnostic: its return correlation with Binance perpetuals is 0.9989 and the mean absolute price spread is 1.98 basis points. We use the perpetual rather than spot because crypto leverage lives in the perpetual, matching a study of the leverage cycle. BTC spot returns serve a dual role: they control for the dominant common crypto factor in the main specification, and they form the basis of the orthogonalized placebo shock (Section 4). Funding rate, the perpetual–spot basis, and open interest describe the leverage state of the market; the high-leverage regime indicator $\texttt{oi\_high}_t$ flags hours where open interest exceeds its 80th rolling percentile over 720 hours. Open interest is Bybit-only (the Binance series is a snapshot, not an hourly history), so the leverage regime is measured at a single venue. We treat this as a limitation. Finally, on-chain liquidations are triggered by oracle prices, which are smoothed and partly CEX-anchored, so the trigger price is not identical to the perpetual price being modeled. This oracle decoupling is one mechanical reason liquidation–price feedback can be attenuated (Caldarelli, 2020; Liu et al., 2021).

2.5 The shock variable

The primary shock is the lagged log flow, $L_{t-1} = \ln(1 + \mathrm{liq}_{t-1}^{\mathrm{USD}})$. The log tames a series whose mean-to-median ratio exceeds two orders of magnitude; without it, a handful of stress hours would mechanically dominate every regression. The unit offset maps the roughly three-quarters of zero hours to zero while remaining negligible against the post-filter minimum of \$5. The one-hour lag means that "$h = 0$" throughout the paper denotes a one-hour-ahead prediction, which attenuates but does not eliminate within-bucket simultaneity. The interaction $L_{t-1} \times \texttt{oi\_high}_t$ carries the leverage-cycle amplification hypothesis. Timing conventions, the raw-versus-orthogonalized shock architecture, and the identification logic are in Section 4.

2.6 Descriptive statistics

Table 1 reports the estimation-panel descriptives: return moments for ETH and BTC (Panel A) and the tail-asymmetry and liquidation co-occurrence measures (Panel B) that anchor the stylized facts of the next section. All entries are computed by a single re-executable script from the canonical panel.

2.7 Data limitations

The data limitations flagged above are collected here in one place: (1) hourly aggregation, hence lost within-hour sequencing and short-horizon bidirectionality; (2) MakerDAO de facto absent; (3) no centralized-exchange liquidations; (4) ETH-like collateral includes liquid-staking derivatives but predates and so excludes the liquid-restaking generation, a one-sided under-count concentrated in 2024–2025; (5) open interest and funding measured at a single venue; (6) BTC–ETH co-movement implies the shock is partly a common-crypto stress proxy (Sections 6 and 9); (7) turnover denominators used for scale comparisons are industry estimates, partly inflated by wash trading, and are treated as order-of-magnitude

Table 1: Descriptive statistics, estimation panel (hourly, 2021–2025).

	ETH perp	BTC spot
Panel A: hourly log-returns (%)		
N (hours)	40,582	40,582
Mean	0.0005	0.0009
Std. dev.	0.812	0.609
Skewness	-0.628	-0.369
Excess kurtosis	17.6	15.0
Min	-15.46	-10.92
p_1	-2.46	-1.86
p_{99}	2.33	1.80
Max	9.17	6.24
Panel B: tail asymmetry and liquidation co-occurrence (ETH)		
$\#\{r \leq -5\%\}$ / $\#\{r \geq +5\%\}$	50 / 25	
$\#\{r \leq -7\%\}$ / $\#\{r \geq +7\%\}$	13 / 6	
$\|p_1\|/p_{99}$ (quantile ratio)	1.05	
$P(\text{liq} > 0 \mid \text{worst 1\% hours})$	85.5%	
$P(\text{liq} > 0 \mid \text{best 1\% hours})$	41.6%	
$P(\text{liq} > 0)$ (all hours)	27.0%	

Notes: Hourly log returns in percent; $N = 40\,582$ (41 328 raw hours less a 746-hour rolling-window warmup). Far-extreme counts are inclusive threshold counts on returns in percent. The co-occurrence rows condition on the within-sample 1% return quantiles and use contemporaneous liquidations ($\text{liq}_t > 0$). Source: `descriptive_stats.csv`, generated by `run_descriptive_stats.py`.

bounds only, since the liquidation numerator is the only hard number; (8) the window is post–DeFi-Summer; (9) the dust and exploit filters make the measure conservative.

3 Three Stylized Facts

Before any estimation, three descriptive regularities frame the question. Together they make the leverage-cycle reading a priori doubtful, which is why the apparently confirming result of Section 5 deserves the scrutiny it receives in Section 6.

Fact 1. ETH's distribution is downside-asymmetric and fat-tailed, but the asymmetry is not specific to ETH. Hourly log returns have skewness -0.63 and excess kurtosis 17.6 (Table 1); the worst hour (-15.5%) is 1.7 times deeper than the best ($+9.2\%$); and far-extreme counts are left-heavy, with 50 hours at or below -5% against 25 at or above $+5\%$ (13 versus 6 at $\pm 7\%$). The measurement deserves precision, because the 1% and 5% *quantiles* themselves are nearly symmetric ($|p_1|/p_{99} = 1.05$): the left-heaviness is a property of the far

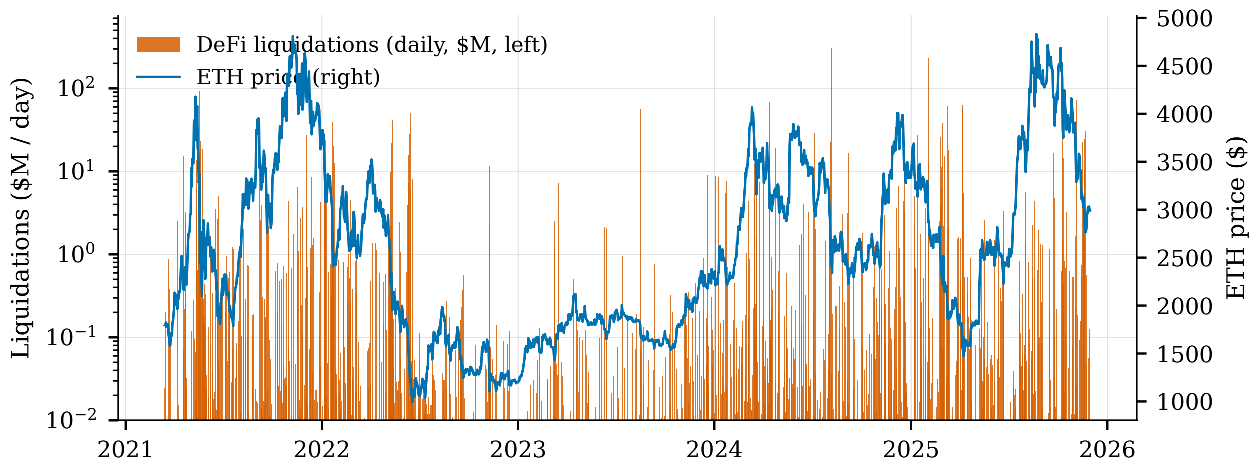

Figure 1: Daily DeFi liquidations (ETH-like collateral, log scale) and the ETH price, 2021–2025. The flow is sparse, right-skewed, and concentrated in a handful of stress episodes.

extremes, not of the moderate tails. Heavy tails as such are the expected stylized fact (Cont, 2001; Osterrieder and Lorenz, 2017), and the *sign* of crypto skewness varies with the return definition and sample: Liu and Tsyvinski (2021) report positive skewness in simple returns across major cryptocurrencies, whereas the hourly log returns studied here skew negative. Two further observations discipline the interpretation. Negative jumps, a driver of volatility documented across the major crypto markets (Ftiti et al., 2021), are not specific to ETH, and tail dependence *across* cryptocurrencies is, if anything, stronger on the upside (Nguyen et al., 2020). So nothing in Fact 1 licenses an attribution of the asymmetry to DeFi. Figure 2 displays the distribution.

Fact 2. Liquidations are a mechanical marker of crashes, not evidence of causing them. Among the worst 1% of return hours, 85.5% contain contemporaneous liquidations, against 41.6% of the best 1% and 27.0% of all hours (Table 1, Panel B). The direction of this co-occurrence is mechanically determined. A price fall pushes positions below their collateral threshold and triggers liquidation, which is the protocol design (Qin et al., 2021; Perez et al., 2021), whereas a rally does not trigger the collateral liquidations we measure. This asymmetry therefore reflects the one-sidedness of the liquidation trigger, not downside amplification of prices by DeFi. Consistently, Moallemi and Patange (2024) document that over 70% of liquidations occur in the absence of any downward price jump, and pool-level price impacts of liquidations are small and temporary (Tian and Zhu, 2025). Fact 2 is the foundation of the paper's register: at the hourly grain the liquidation–return relation is bidirectional at short horizons, so every estimate below is read as predictive association, never as a causal effect.

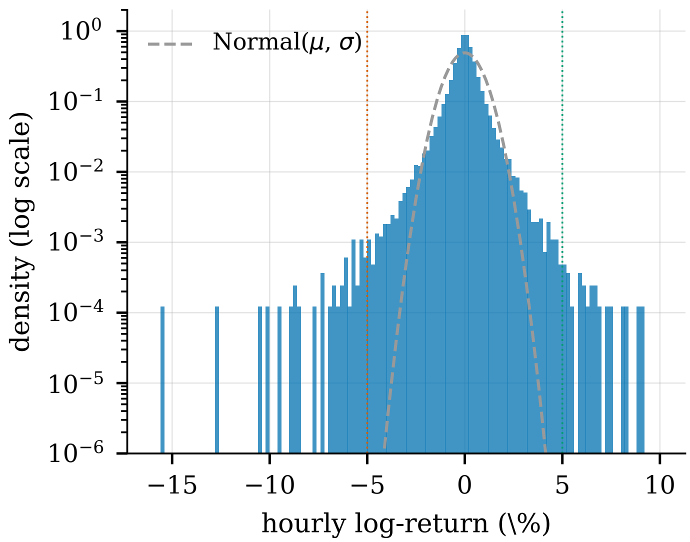

Figure 2: Hourly ETH log returns (density, log scale) against a normal benchmark. Dotted lines mark the ±5% far-extreme thresholds of Table 1.

Fact 3. The observable flow is minute relative to the ETH market. Median daily DeFi liquidations amount to roughly 2×10^{-5} of one percent of estimated ETH spot-plus-perpetual turnover; the mean is 0.0032%; and even the worst liquidation day of the sample reaches only 0.67% (Table 7 below reports the full set). Against the largest reported centralized-exchange liquidation cascade on record, roughly nineteen billion dollars in a single day (CoinGlass, 2026), the typical daily flow sits four to six orders of magnitude lower. The denominators are industry estimates, partly inflated by wash trading and not peer-reviewed, so these are order-of-magnitude bounds. But the inequality is wide enough to survive any plausible rescaling, and the numerator is our own hard extraction. Fact 3 pre-loads the mechanism discussion of Section 8: whatever the within-pool impact of an individual liquidation, the aggregate observable flow is small relative to the market that would need to be moved.

These three regularities, a real but cross-asset asymmetry, mechanically reverse co-occurrence, and negligible scale, make it doubtful from the outset that observable DeFi liquidations drive ETH's downside tail. The remainder of the paper tests that doubt formally: the next section fixes the specification, Section 5 shows how a naive quantile analysis nevertheless appears to confirm the leverage-cycle prediction, and Section 6 explains why that appearance is an artifact of interpretation.

4 Empirical Strategy

4.1 The estimand

For each quantile τ and horizon h we estimate, by local projection (Jordà, 2005), the response of the conditional return quantile to the liquidation shock:

$$\mathrm{Q}_\tau(r_{t+1\to t+1+h} \mid \mathcal{F}_t) = \beta_h(\tau)\, L_{t-1} + \delta_h(\tau)\ (L_{t-1} \times \texttt{oi\_high}_t) + \gamma_h(\tau)' X_t, \qquad (1)$$

where $r_{t+1\to t+1+h}$ is the cumulative ETH return over the $h+1$ hours from $t+1$ to $t+1+h$ (so $h = 0$ is the one-hour-ahead return), L_{t-1} the lagged log-liquidation shock, and X_t the controls of Section 4.3. The object of interest is $\beta_h(\tau)$ across $\tau \in \{0.01, 0.05, 0.10, 0.50, 0.90, 0.95\}$ and $h \in \{0, \dots, 24\}$ hours (a nine-quantile grid is reported in the appendix). Estimation is by quantile regression (Koenker and Bassett, 1978); the estimator and its bootstrap inference follow Han et al. (2024). The reported grid stops at $\tau = 0.95$; the tail-inference and mirror set additionally estimates $\tau = 0.99$, so that the one-percent upper tail has a symmetric counterpart to the one-percent lower tail. Both grids are fixed in the locked configuration file, so 0.99 is estimated but not carried into the reported panels.

The quantile approach is necessary rather than cosmetic, for two reasons. First, the hypothesis under test, leverage-cycle amplification (Geanakoplos, 2010; Brunnermeier and Pedersen, 2009), is a statement about the *tails* of the return distribution, not its mean. Second, the mean is empirically silent here: an OLS local projection of returns on the shock finds nothing at any horizon (point estimate ≈ -0.0001 on impact, HAC $p = 0.94$). This is the Growth-at-Risk logic of Adrian et al. (2019) transposed to high-frequency crypto returns: tails can respond where the mean is blind. We note at the outset a tension that Section 6 resolves. Adrian et al. (2019) interpret asymmetric tail sensitivity through conditional skewness; Carriero et al. (2024) show that symmetric conditional distributions, under joint shifts in mean and variance, can produce the same apparent pattern; and Brownlees and Souza (2021) find that a plain GARCH rivals quantile-regression Growth-at-Risk out of sample. Distinguishing those readings is the heart of this paper.

4.2 Shock definition and timing

The canonical shock is *raw*: $L_{t-1} = \ln(1 + \mathrm{liq}^{\mathrm{USD}}_{t-1})$, one-hour lagged (Section 2). Timing is read strictly. "$h = 0$" is a one-hour-ahead prediction; within the bucket at $h = -1$, prices and liquidations co-move mechanically (the reverse-causality zone of Fact 2); only $h \leq -2$ would be a true lead. All claims are therefore predictive associations.

A second, *orthogonalized* shock (the residual of an OLS regression of the log flow on contemporaneous and lagged BTC returns, then lagged) is reserved for the cross-asset placebo (Test A of the robustness battery), in the ex-ante residualization spirit of Borusyak et al. (2022). The two definitions preserve sign and significance at every horizon and agree to within a few percent from six hours out, but diverge at short horizons (the impact tail coefficient is about 60% larger under orthogonalization). The choice between raw and orthogonalized is thus an identification convention that leaves the symmetry finding and the bound intact, not a claim of numerical coincidence at every horizon; residualizing at the quantile (rather than by OLS) is an acknowledged refinement left for future work.

4.3 Controls and the identification logic

The regressor set is locked: $\{L_{t-1},\ L_{t-1} \times \texttt{oi\_high}_t,\ \texttt{oi\_high}_t,\ r_t^{\text{BTC}},\ \sigma_t^{7d},\ \text{funding}_t,\ \text{basis}_t\}$. The realized-volatility control σ_t^{7d} (168-hour rolling standard deviation) is not a nuisance term: it is central to the identification. Liquidations raise volatility, and volatility widens *both* tails symmetrically; the leverage-cycle prediction is specifically *downside* amplification. Holding σ_t^{7d} fixed conditions $\beta_h(\tau)$ on the *predetermined* volatility state, removing the confound between the shock and the pre-existing volatility regime. It does not remove the shock's own forward-volatility response (Section 7.1). But that response is symmetric, so it cannot manufacture asymmetry; it is the symmetric residual that the placebo and the paired tail tests identify directly. The asymmetry tests, not this control alone, are therefore what separate a real downside shape from a symmetric scale effect. That a symmetric conditional distribution can masquerade as downside risk is the demonstration of Carriero et al. (2024), there through joint shifts in mean and variance; the scale-versus-shape distinction is what quantile regression makes testable (Koenker and Bassett, 1982; Machado and Silva, 2000). Realized volatility is preferred to a GARCH or implied measure because it is observable, model-free, and predetermined; model-based volatility enters only inside the placebo DGPs of Section 6, where its model-dependence is the point.

4.4 Inference

Inference proceeds on two tiers. At central quantiles, kernel standard errors (Hall–Sheather bandwidth) are retained. At tail quantiles ($\tau \in \{0.01, 0.05, 0.95, 0.99\}$), the primary inferential object is the percentile confidence interval from a moving-block bootstrap with 24-hour blocks (Fitzenberger, 1998; Politis and Romano, 1994; Gregory et al., 2018), for two reasons. First, the cumulative left-hand side is overlapping by construction, so residuals are serially correlated at every $h > 0$ (Montiel Olea and Plagborg-Møller, 2021; Lusompa, 2023); blocks of one day, the maximal horizon, preserve that dependence. Second, at $\tau = 0.01$ roughly 406 observations sit below the conditional quantile, and the extremal quantile regression literature shows the asymptotic distribution is non-Gaussian that far in the tail (Chernozhukov, 2005; Chernozhukov and Fernandez-Val, 2011), making kernel inference unreliable: on the exact headline specification, the bootstrap-to-kernel standard-error ratio at $\tau = 0.01$ ranges from 1.6 to 3.3 across horizons (Appendix Table 8). All bootstrap results use 1,000 replications with a deterministic multi-level seeding scheme. The MBB block length does not drive the results: re-running the tail inference at 12-, 36-, and 48-hour blocks leaves every interval and every minimum detectable effect essentially unchanged (Appendix Table 9).

4.5 The diagnostic battery

The core methodological contribution is a battery of diagnostics. Each one is designed to isolate a single confound of the naive reading. What survives all of them is what we are willing to call a real signal.

Table 2 summarizes the battery; applications and results are in Sections 6–7. The equivalence step requires a pre-specified smallest effect size of interest (SESOI). We set it at a one-percentage-point difference in tail-violation probability between the downside and upside over the span of the shock, translated into coefficient units under three calibrations of that span (interquartile, median-to-95th, and 10th-to-90th percentile of the nonzero shock); the strictest yields 0.0011 per unit of the log shock. The one-point threshold is decisional, not statistical. It is the smallest down-versus-up tail miscalibration a Value-at-Risk desk would act on, not the smallest the sample can resolve (Lakens, 2017). One percentage point is the order at which a shift in the realized 1% exceedance rate moves a model across the Basel traffic-light boundary over the standard 250-day backtesting window. That window is the institutional unit at which tail calibration is judged by mandate, and we anchor the threshold to it rather than to the far finer statistical resolution of a forty-thousand-hour sample.

4.6 Multiple testing and the locked specification

Bonferroni corrections guard the grid searches: a $\times 12$ family over the reported $\tau \times h$ cells and a $\times 6$ family over the quantile grid. The correction family is the set of reported cells itself, the $\tau \times h$ grid and the quantile grid enumerated in the locked configuration file before the diagnostic phase, so the family is fixed by the specification rather than chosen after seeing the diagnostics. The direct shock effect survives both families in full (12/12 and 6/6); the leverage interaction $\delta_h(\tau)$ fails all six (0/6). We carry that null forward as it stands, since the interaction was the specification's embodiment of the leverage-cycle prediction. We read it as a *detectability* null, not a tight bound (the interaction is imprecisely estimated; Section 5.4), and the paper's quantitative bounds rest on the asymmetry objects, not on this count. The corrections guard against noise, not against artifacts, which is exactly why the battery of Table 2 exists.

The specification was locked before the diagnostic phase: every parameter (quantile grids, horizons, controls, block length, replication counts, seeds) is fixed in a single configuration file, and every diagnostic is a flagged auxiliary analysis, never a modification of Equation (1). The full pipeline, from panel construction through estimation, robustness, diagnostics, and every number, table, and figure in this paper, regenerates from public data with a one-command replication package; local smoke runs and the canonical compute environment reproduce point estimates to three decimals.

5 The Apparent Result

5.1 The naive quantile profile

Table 3 and Figure 3 report the headline estimates of Equation (1). On impact, the response of the first percentile is $\beta_0(0.01) = -0.032$, strongly significant under the tail-appropriate block-bootstrap inference, while the median barely moves ($\beta_0(0.50) = 0.0017$). The tail response is 18.8 times the median response. The left tail also appears to respond more than the right, with $|\beta_0(0.01)|/|\beta_0(0.95)| = 1.75$ on impact.

At first reading this is what the leverage-cycle prior[2] predicts (Geanakoplos, 2010; Brunnermeier and Pedersen, 2009): a liquidation shock depresses the crash quantile of returns far more than the center, and more than the upside. It is, term for term, the Growth-at-Risk pattern of Adrian et al. (2019), with the lower tail responding where the mean is blind (our OLS control regression sees nothing at any horizon). And it is produced by an established estimator (Han et al., 2024), not an exotic one. The reservation entered below is never about the *existence* of β; it is about its *interpretation*.

[2]The view we deconstruct is actively held, not a strawman built for contrast. Alongside the leverage-cycle theory cited here it rests on the fire-sale tradition (Shleifer and Vishny, 1992), finds empirical grounding for DeFi in the leveraging spirals documented by Perez et al. (2021) and the systemic fragility analyzed by Lehar and Parlour (2022), and is monitored by the official sector as a financial-stability concern (Sasi-Brodesky and Nassr, 2023; Financial Stability Board, 2023).

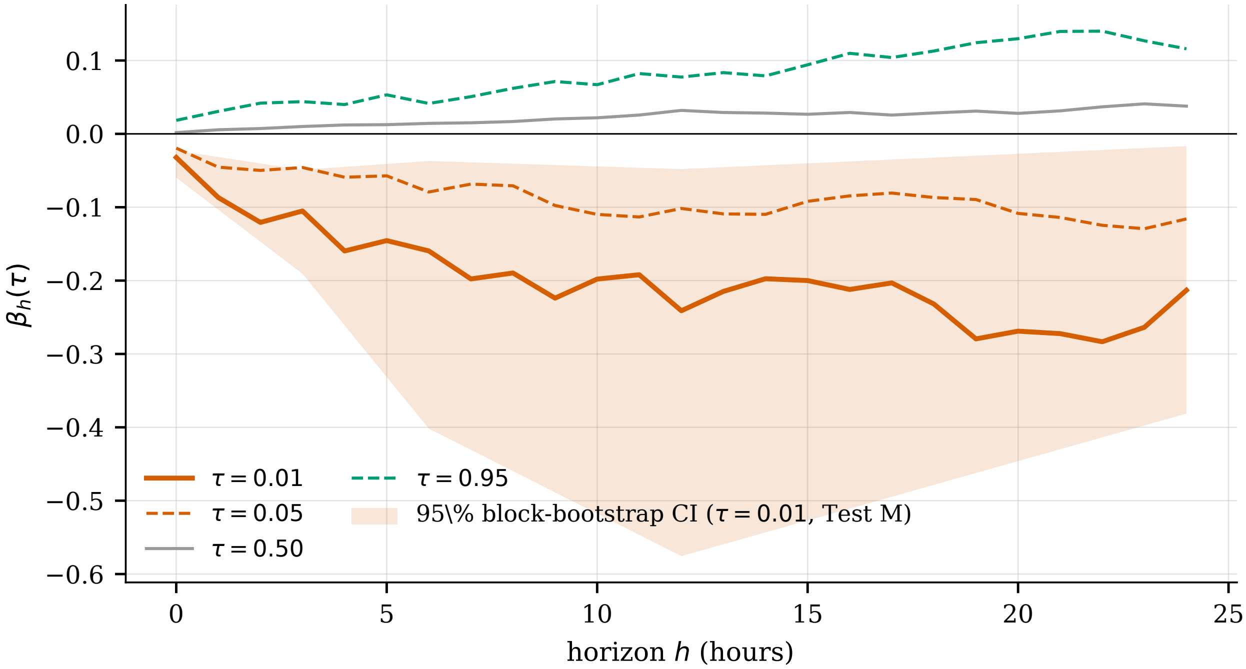

Figure 3: Quantile local-projection responses $\beta_h(\tau)$ with the 95% block-bootstrap band on $\tau = 0.01$ (headline specification). The fan widens and the lower tail deepens with the horizon.

5.2 The deepening profile

The tail coefficient deepens almost monotonically with the horizon: -0.032 on impact, -0.087 at one hour, -0.241 at twelve, with a plateau near -0.213 at twenty-four (Table 3). An effect that grows as the window lengthens reads naturally as the footprint of a spiral, the shock propagating and self-reinforcing, which is the reading a leverage-cycle prior invites. We flag, without yet arguing, that the left-hand side at $h > 0$ is an overlapping cumulative return, a construction whose tail behavior under the null deserves dedicated scrutiny, and receives it in Section 6 (Hodrick, 1992; Boudoukh et al., 2008; Neuberger and Payne, 2021; Farago and Hjalmarsson, 2023).

5.3 The apparent robustness

The pattern checks every box a referee would ask for. The block-bootstrap band on $\tau = 0.01$ excludes zero at every horizon (on impact $[-0.060, -0.023]$; at twelve hours $[-0.576, -0.048]$; at twenty-four $[-0.381, -0.017]$; 1,000 replications). The sign and significance survive a battery of conventional robustness checks, including placebo assets, alternative shock definitions, subperiod and episode exclusions, and monotonicity across quantiles, reported in Appendix A. And the direct effect survives Bonferroni correction in full: 12 of 12 cells of the reported $\tau \times h$ grid and 6 of 6 of the quantile grid. By the usual standards of empirical asset pricing, this is a robust result.

Two cautions temper the checklist. Multiple-testing corrections guard against *noise*, not against *misinterpretation*. Surviving Bonferroni says the effect is not sampling luck; it does not say the effect is a downside skew rather than a symmetric volatility response read through extreme quantiles. And at $\tau = 0.01$ the extremal quantile literature counsels explicit caution on point estimates (Chernozhukov, 2005; Chernozhukov and Fernandez-Val, 2011); tail inference here therefore rests on interval estimates, from the moving-block bootstrap motivated on separate grounds in Section 4.4.

5.4 The leverage interaction

The leverage-cycle story makes a sharper prediction than "liquidations depress the tail": amplification should be *stronger when leverage is high*. The interaction term $\delta_h(\tau)$, the specification's direct embodiment of that prediction, is null across the tail-quantile grid. At $\tau = 0.01$ the bootstrap interval straddles zero at every horizon (on impact $[-0.080, 0.072]$; at twelve hours $[-0.392, 0.458]$), the sign is unstable across resamples, and the Bonferroni family verdict is 0 of 6. This is not a nuance to be excused. The one mechanism-specific term in the regression fails on the tail grid, while the mechanism-agnostic direct term looks spectacular. The leverage-cycle reading already has a hole in it.

We are careful about what this null does and does not establish. Unlike the direct effect, the interaction is estimated with little precision: the bootstrap interval is wide and its sign flips across resamples, so the smallest leverage amplification it could detect at conventional power, 0.109 at impact, is a multiple of the direct effect itself, roughly 2.1 to 3.4 times it, not a fraction of it. The 0/6 verdict is therefore best read as the *absence of any detectable* leverage-specific amplification rather than as a tight bound on its magnitude, a hole in the mechanism story rather than a measured zero. The decisive, quantified bounds in this paper attach to the symmetric-scale and down-minus-up asymmetry objects (Section 7), where standardizing by predetermined volatility cancels the common scale and sharpens the test; the interaction inherits no such cancellation, which is why it stays wide. The interaction is, however, not null at the centre of the distribution: at the median it is significantly negative at the twelve- to twenty-four-hour horizons ($\delta_{12}(0.50) = -0.024$, $p =< 0.001$), so high-leverage hours shift the shock's median response downward. That is a level effect on the centre of the return distribution, not the downside-tail amplification the leverage-cycle prior predicts, and it is orthogonal to the down-minus-up asymmetry this paper bounds; it is consistent with a liquidation channel that is real and locally disruptive yet does not aggregate into tail fragility (Section 8).

5.5 Summary of the apparent result

Taken together: a direct effect 18.8× the median on impact, deepening to -0.241 by twelve hours, robust across a battery of robustness checks, surviving multiple-testing correction in full, and a mechanism-specific interaction not detectably different from zero, imprecisely estimated rather than tightly bounded, with the paper's decisive, quantified bounds attaching instead to the symmetric-scale and down-minus-up asymmetry objects (Section 7). The question that the next section answers is the one this contrast forces. Is the real, robust direct effect a genuine *downside* response, or the same symmetric volatility that thickens *both* tails, misread by a quantile local projection on overlapping cumulative returns?

6 Deconstruction

6.1 What must be explained

Section 5 leaves three facts to explain: a strong negative $\beta_h(0.01)$ that deepens from -0.032 to -0.241 over twelve hours; a tail-to-median ratio of 18.8 on impact; and a left tail that appears to respond more than the right. Three readings compete. The first is genuine downside-specific amplification, the leverage-cycle interpretation; the second is a genuine but *symmetric* volatility channel, misread; the third is a mechanical artifact of the overlapping cumulative design. We do not adjudicate by assertion. Each alternative is confronted with a test built to manufacture the stylized fact under a known null: if the test reproduces the fact, the fact is not a signature of the first reading. Two distinctions organize the section. The *level* of β and the *shape* of the response are separate questions. Level means the magnitude of $|\beta_h(0.01)|$ (the scale channel of Section 4.3, tested by the permutation nulls); shape means the left-minus-right gap $g_h = |\beta_h(0.01)| - |\beta_h(0.99)|$[3] (tested by the placebo). A coefficient can therefore be genuine in level yet symmetric in shape, and that, we will conclude, is exactly what the data say. And ETH-specificity is a separate question again, tested by replacing the outcome altogether.

6.2 The symmetric placebo

We simulate a world that is symmetric by construction and re-run the exact estimation kernel on it. Returns are rebuilt as $r_t^{\text{sim}} = m_t \pm |\hat{\varepsilon}_t|$, where m_t is the fitted conditional mean

[3] The gap pairs each tail quantile with its equidistant mirror: 0.01 with 0.99, and the headline ratio of Section 5 pairs 0.01 with 0.95. The reporting grid stops at 0.95, but 0.99 is estimated as part of the tail-inference set (Section 4.4); the headline ratio uses 0.95 as the rhetorical object, while the rigorous left-versus-right test below uses the true mirror 0.99.

and the sign is an independent Rademacher draw, the sign-flip construction of the wild-bootstrap literature (Wu, 1986; Liu, 1988; Davidson and Flachaire, 2008). The simulated world thus has *zero skew by construction* at every point in time, while preserving the empirical magnitude sequence exactly: the full fat-tailed, volatility-clustered path of $|\hat{\varepsilon}_t|$, with no volatility model imposed at all. Because the sign-flip removes asymmetry while leaving the volatility path untouched, any left–right gap this world produces is, by construction, what symmetric heteroskedasticity alone manufactures through extreme quantiles. A gap that the real data merely matches therefore cannot be evidence of real downside skew. Only the dependent variable is simulated; the shock and all controls are the real series. Each of 500 simulated panels is passed through the identical quantile-LP kernel, and the statistic of interest is the left–right gap $g_h = |\beta_h(0.01)| - |\beta_h(0.99)|$. The level of $|\beta|$ is expected to survive (the volatility path is real, which is the point), so the test bites on the *asymmetry*, which the design rules out as a property of the data.

The zero-skew world reproduces the observed asymmetry everywhere (Figure 4). At twelve hours the real gap (0.153) sits near the center of the placebo distribution ($[-0.015, 0.343]$); on impact the real gap (0.014) lies within the symmetric band ($[-0.019, 0.018]$). There is no real impact gap either. Nor is this verdict an artifact of the band's width. Sweep the placebo's tail thickness across a variance-preserving family, from thin- to heavy-tailed, and the real gap stays inside the symmetric band at the data's own tail thickness and under every heavier tail; it escapes only when the simulated world is forced thinner-tailed than the data, an artificially narrow band that under-represents the spread the returns actually carry. A dose-response sweep of the heteroskedasticity *amplitude* (tracing the manufactured gap against a tail-thickness parameter) would only confirm the monotonicity of a mechanism the placebo already exhibits at the data's own heteroskedasticity, at the cost of an arbitrary parametric family, and is omitted. A vol-model-scaled variant, in which standardized magnitudes are permuted across dates and rescaled by a fitted volatility path (a shock-driven rolling specification and a shock-agnostic GARCH(1,1), two distinct designs), delivers the same verdict in all fourteen model-horizon cells. The observed left–right asymmetry is therefore what symmetric heteroskedasticity looks like through extreme quantiles of cumulative returns, the mechanism demonstrated for macroeconomic tail risk by Carriero et al. (2024), instantiated here on asset returns. Consistently, Brownlees and Souza (2021) find that a symmetric GARCH rivals quantile-regression tail-risk models out of sample: the scale channel does the work. One reading discipline applies. Where the placebo occasionally generates more asymmetry than observed, the correct statement is that the real gap does not exceed what a zero-skew world generates; the test is one of non-exceedance, not of equality.

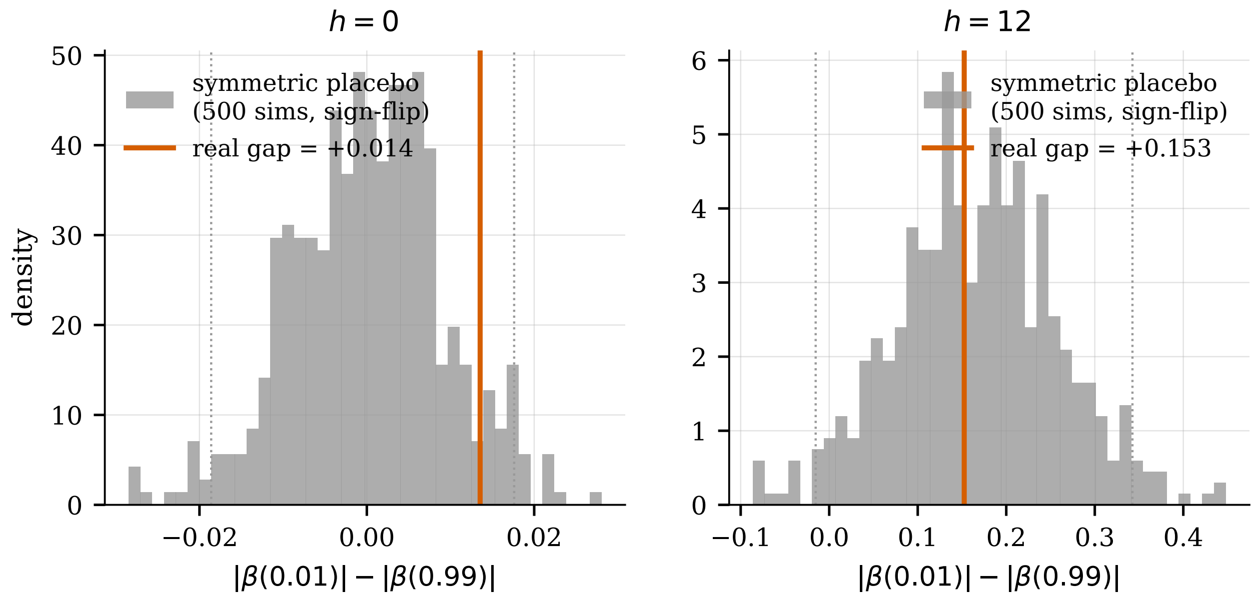

Figure 4: Distribution of the left–right tail gap $|\beta(0.01)| - |\beta(0.99)|$ across 500 zero-skew sign-flip placebo worlds (grey; dotted lines mark the 2.5th/97.5th percentiles), against the real gap (vertical line), on impact and at twelve hours. The real asymmetry is indistinguishable from what a symmetric world produces.

6.3 The permutation nulls

The placebo addresses shape. The level question, whether the deepening $\beta_h(0.01)$ is a mechanical artifact of the overlapping cumulative design (Hodrick, 1992; Boudoukh et al., 2008; Valkanov, 2003), is answered with two permutation nulls run at 500 draws each, and reported jointly as a guard against null-shopping. The conservative null circularly shifts the shock series, preserving its marginal distribution and autocorrelation (and the real, heteroskedastic dependent variable) while severing only its alignment with returns. The coarser null reshuffles the return innovations themselves, destroying their serial dependence; the two designs bracket what a “no information” world can mean here, and diagnostics of what each null preserves and destroys are recorded with the artifacts.

Both nulls return the same verdict (Figure 5). Mechanical noise accounts for a minority of the observed coefficient at every horizon: the null-to-real magnitude ratio at twelve hours is 0.18 under the conservative null and 0.08 under the coarser one, rising to at most 0.35 at twenty-four hours. The signed mean of the null distribution is essentially zero throughout (e.g. 0.004 at twelve hours, against a real coefficient of -0.241): the overlap inflates the *dispersion* of tail estimates, it does not push them downward. And the real coefficient lies outside the null band ($[-0.12, 0.10]$ at twelve hours) at every horizon, with permutation p-values below

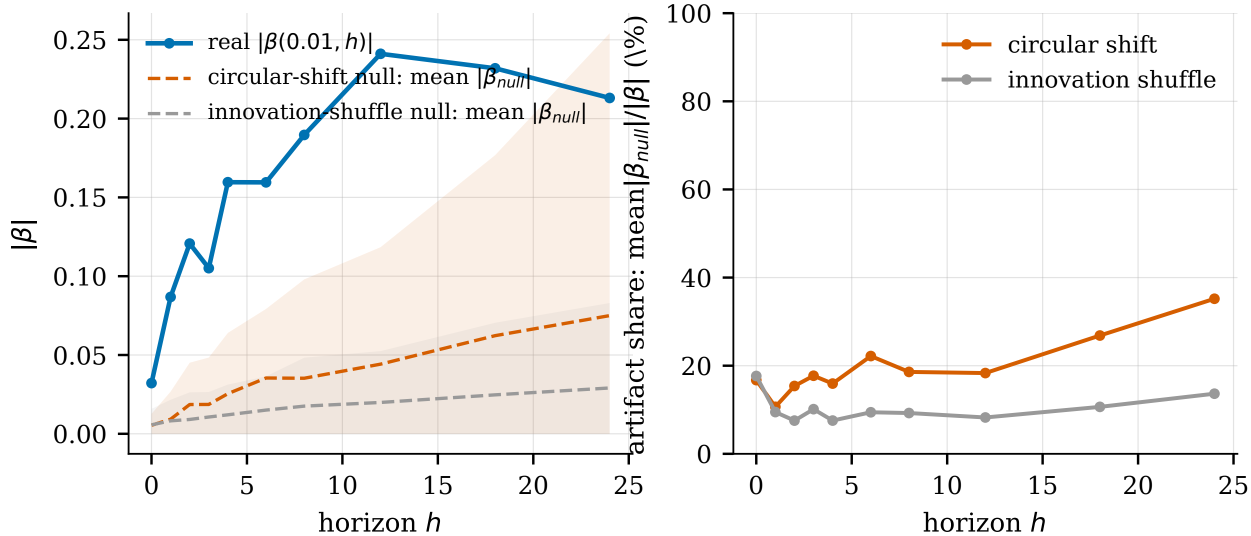

Figure 5: Dual permutation nulls. Left: the real $|\beta_h(0.01)|$ against the mean null magnitude under each construction (shaded: null bands). Right: the null-to-real ratio by horizon, a minority share everywhere, under both nulls.

0.005 through eighteen hours and .038 at twenty-four. The level is overwhelmingly real.[4] This matters in both directions. It removes the artifact reading as a primary explanation, and it sharpens the burden on interpretation: the coefficient is real, so the question is entirely what it means.

6.4 The overlap artifact

If the overlapping cumulative construction misleads about *shape* rather than level, the cleanest demonstration is to remove it. The exceedance regressions of Section 7 use per-period (non-overlapping) tail-violation indicators; their paired asymmetry $\Delta_h = \beta_h^{\text{down}} - \beta_h^{\text{up}}$ is null at every horizon (at twelve hours, $p = .844$). Re-running the same construction on overlapping cumulative returns manufactures a significant asymmetry at exactly the horizons where the naive reading is most dramatic ($\Delta_{12} = -0.0044$, $p = .023$; Figure 7, right panel). Same data, same controls, same estimator family. The only difference is the overlap. The horizon-dependent skewness that cumulation fabricates (Neuberger and Payne, 2021; Farago and Hjalmarsson, 2023) is not a theoretical curiosity here; it is visible in one comparison.

One might worry the null is an artifact of conditioning, that holding the controls fixed is what suppresses the asymmetry. Re-estimating the effect on the *marginal* return quantile, by the unconditional-quantile (RIF) method, answers it. On the density-normalized

[4] The medium-horizon *magnitude* is separately sensitive to the MakerDAO coverage gap (Appendix C); the permutation nulls certify only that the deepening is not a mechanical artifact of overlap, a distinct question from coverage.

scale the combined tail-widening at impact is large (-0.005), both tails opening together, the volatility channel again, while the down-versus-up asymmetry recovered from the same estimates reproduces the conditional exceedance ($\Delta_0 = 0.00053$) to machine precision at every horizon. The agreement is partly algebraic: RIF-OLS with these controls is the conditional linear-probability model up to the density factor, so it confirms the null rather than re-deriving it independently. What the unconditional lens adds is the one distinct object, the large symmetric widening, and it points the same way.

6.5 The headline ratio

The "18.8×" headline deserves a closer look, because it is rhetorically the strongest number in Section 5 yet statistically the least informative. Volatility widens both tails but does not move the center: $\beta_h(0.50) \approx 0$ is what a scale channel implies (and what the silent OLS mean regression already said). A tail-to-median ratio with a near-zero denominator explodes mechanically; it measures "the median is immobile," not "the downside is special." Diverging quantile slopes, the fanning of Figure 3, are the textbook signature of heteroskedasticity, not of shape change (Koenker and Bassett, 1982; Machado and Silva, 2000). And the correct contrast, left tail against right tail, is the one the ratio hides: $\beta_{12}(0.05) = -0.102$ against $\beta_{12}(0.95) = 0.077$, both tails moving. At horizons beyond impact the median itself drifts positive ($\beta_{12}(0.50) = 0.032$), and the ratio collapses to 7.5×. The headline number is also impact-specific.

6.6 The BTC-outcome placebo

A final alternative survives the tests above: perhaps the response is real and symmetric, and still specifically about ETH's DeFi collateral channel. The sharpest test replaces the outcome. We re-run the full quantile local projection, identical shock and horizons, with Bitcoin returns as the dependent variable (the cross-asset control becoming ETH's return, by symmetry; BTC has no direct ETH-collateral forced-selling channel, yet is a highly correlated benchmark, so a replicated signature demonstrates non-specificity rather than a null effect). The ETH signature replicates on Bitcoin in every respect (Figure 6): $\beta_0^{\mathrm{BTC}}(0.01) = -0.023$ on impact, deepening to -0.145 at twelve hours, a near-parallel profile at 0.70 of the ETH magnitude on impact and 0.60 at twelve hours, with an even more extreme tail-to-median ratio on impact (46.8×, BTC's median response being essentially zero). The naive signature is therefore not evidence of an ETH-specific DeFi channel. It is what generic crypto stress looks like through this lens, on any correlated asset. The residual claim of ETH specificity is accordingly demoted: BTC carries most of the signature, and no ETH-specific increment

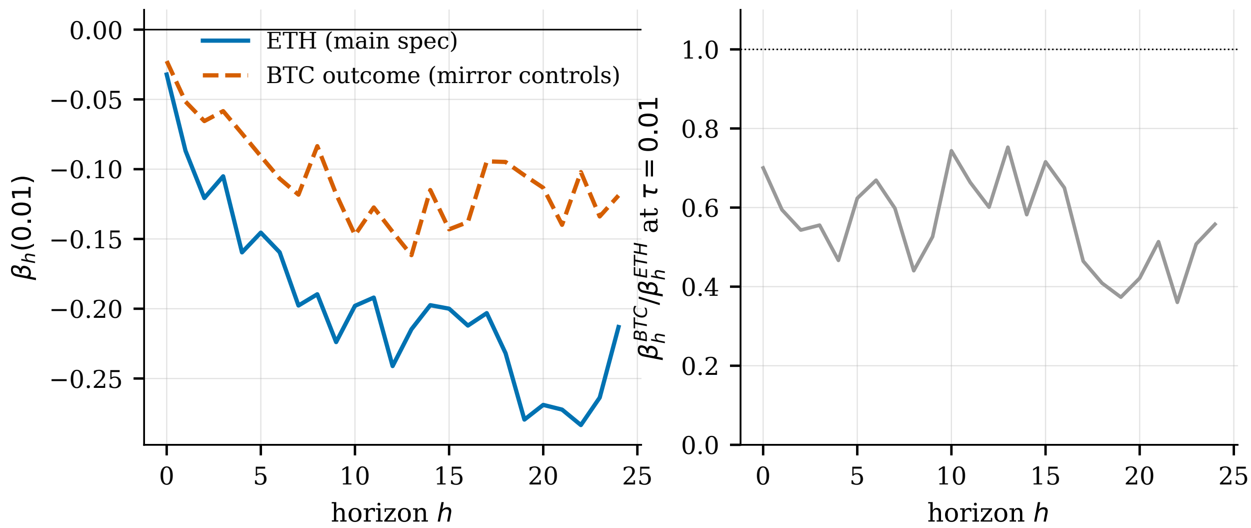

Figure 6: BTC-outcome placebo. Left: $\beta_h(0.01)$ profiles, ETH (main specification) and BTC as outcome (mirror controls). Right: the BTC-to-ETH coefficient ratio by horizon. The apparent "downside-amplification" signature replicates on an asset without the ETH-collateral channel.

large enough to drive the result survives. The methodological lesson gains generality, since the misreading would afflict any asset regressed on any stress-correlated flow.

6.7 The portable detection battery

What is new here, and what is borrowed, should be stated carefully. The mechanism, symmetric and time-varying volatility producing apparent downside sensitivity in quantile models, is established in the macroeconomic Growth-at-Risk setting by Carriero et al. (2024); the overlap pathologies of long-horizon regressions are classic (Hodrick, 1992; Boudoukh et al., 2008; Valkanov, 2003); scale-versus-shape in quantile regression goes back to Koenker and Bassett (1982) and Machado and Silva (2000). What is new, to our knowledge, is the *conjunction* in which all four ingredients meet (asset returns, overlapping cumulation, quantile local projections, and the extreme tail) together with a diagnostic battery that detects the trap on real data: a model-free zero-skew placebo for shape, dual permutation nulls for level, a per-period-versus-cumulative comparison that exhibits the artifact directly, a cross-asset outcome placebo for specificity, and a planted-cascade positive control under which the equivalence apparatus recovers a downside asymmetry at twice the actionable threshold and stays silent below it, so the bounded null it returns is power-confirmed, not power-absent. The lesson travels: any literature that regresses extreme quantiles of cumulative returns on stress-correlated regressors, and the nascent empirical DeFi literature is about to be one, can produce "endogenous fragility" of this apparent strength from a symmetric volatility channel.

6.8 Reconciliation

The reconciled reading of Section 5 is now complete. In *level*, $\beta_h(0.01)$ is overwhelmingly genuine: both permutation nulls leave mechanical noise a minority share, with no signed bias. In *shape*, the raw QLP asymmetry is not evidence of economically meaningful downside-specific shape. A zero-skew world reproduces the entire left–right gap at every horizon, the per-period construction shows both tails loading equally, and the headline ratio is an artifact of an immobile median. In *specificity*, the signature replicates on Bitcoin. What the naive analysis detected is a real, symmetric, generic-crypto volatility response, the second moment dressed up as downside fragility. What remains to establish is what survives on the positive side, and how tightly the null can be bounded; that is Section 7.

7 What Survives

After the deconstruction, the question is no longer whether the apparent downside amplification is real. As a *downside* effect it is not. The question is what remains once volatility and the overlap artifact are accounted for. Three objects survive the battery, and they tell a single coherent story: a real second-moment channel, a symmetric and out-of-sample-validated tail predictor, and a null that is not merely insignificant but *bounded*.

7.1 The volatility channel

Regressing forward realized volatility over the h-hour window on the shock and the full control set yields positive, strongly significant coefficients at every horizon, rising from 0.021 on impact to 0.065 at twelve hours (per unit of the log-liquidation shock; an absolute-return measure confirms sign and scale). One unit caveat applies throughout. The coefficient is percentage points of volatility per unit of log liquidations, not an elasticity per percent liquidated, so only the sign and order of magnitude are comparable to the transaction-level estimates of Sasi-Brodesky and Nassr (2023), whose direction we replicate and extend from the conditional mean to the conditional tails. This channel is real, expected, and mundane. It is also the engine of the entire deconstruction. It is this symmetric second-moment response that, propagated through cumulative returns and extreme quantiles, masqueraded as downside amplification in Section 5.

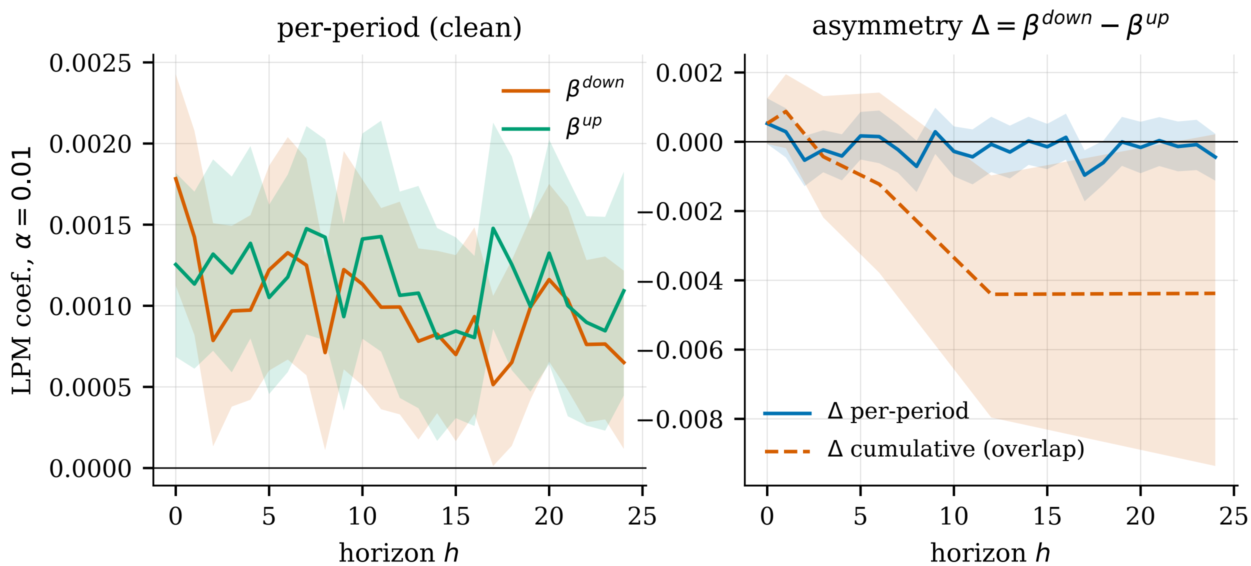

Figure 7: Per-period exceedance coefficients and paired asymmetry. Left: per-period exceedance coefficients at $\alpha = 1\%$, where both tails load positively. Right: the paired asymmetry Δ_h under the per-period construction (null everywhere) and under the overlapping cumulative construction (a significant asymmetry is manufactured at long horizons), the overlap artifact of Section 6.4.

7.2 The tail-risk predictor

In sample, the per-period exceedance regressions (tail-violation indicators in the CAViaR tradition of Engle and Manganelli (2004), estimated with the full control set, conditioning on the predetermined volatility state) show the shock loading positively on *both* tails at every level and horizon (Figure 7, left): on impact at the 1% level, $\beta^{\text{down}} = 0.00178$ and $\beta^{\text{up}} = 0.00125$; at the 5% level, 0.00393 and 0.00359 (all bootstrap intervals exclude zero). Liquidations predict elevated tail risk on both sides of the distribution. This two-sided loading has a natural reading in the microstructure of information: just as a market maker's own quotes can reveal the direction of their inventory to others (the price-reading channel of Barzykin et al. (2025)), an observable, predetermined flow of forced deleveraging is a public signal other agents can read. What it reveals is the magnitude of impending risk rather than its sign, which is why liquidations forecast elevated violations in both tails symmetrically rather than a signed downside amplification. To our knowledge this is among the first quantile-resolution evidence on on-chain forced deleveraging. Adjacent work studies on-chain flows into means and volatilities, or trading volume across return quantiles without finding volatility effects (Balcilar et al., 2017).

An in-sample coefficient, however robust, is not yet a predictor. We use a rolling-origin evaluation (weekly re-estimation, pinball loss on the held-out tail forecasts, Diebold–Mariano inference on the loss differentials (Diebold and Mariano, 1995)). Adding the liquidation terms to an otherwise identical quantile model improves tail forecasts directionally in all twenty nested-QR cells, most not individually distinguishable from zero, and significantly at both moderate tails at short horizons (at one hour: 1.0% at $\tau = 0.05$, $p = .004$; 0.9% at $\tau = 0.95$, $p =< 0.001$; Table 4). These two cells, and a third at a long horizon ($\tau = 0.95$, twenty-four hours), survive a Benjamini–Hochberg correction across the grid (3 of the twenty do, FDR-adjusted $p = .037$ and .014; the 95% cell survives even the stricter Holm bound), so the surviving positive finding meets the multiple-testing discipline applied to the interaction null. The symmetric reading carries out of sample. The boundary is equally clear. Against a dedicated GARCH(1,1) scale benchmark the picture is deliberately different: the augmented model does not systematically outperform it, the short-horizon cells that beat the nested benchmark do *not* beat GARCH (one, $\tau = 0.95$ at one hour, significantly underperforms it), and the only cells that beat GARCH are sparse long-horizon ones. That boundary is the point rather than a concession. The liquidation signal is observable, predetermined, and external to the price-only forecasting model, so the information it carries is orthogonal and incremental to the market-state controls a risk desk already tracks: a free input to add to an existing tail-risk model, not a replacement for one. Because pinball loss is the objective a quantile VaR forecaster minimizes, the one-percent reduction at the moderate tails reads directly as a calibration gain in a one-hour-ahead, on-chain-observable VaR signal, consistent with the Brownlees and Souza (2021) finding that symmetric scale models are formidable tail forecasters. A dollar quantification of that gain would require position-level data and is left for future work.

7.3 The bounded null

Net of volatility, downside-specific amplification is not economically significant, and what is excluded is quantified in three steps. This down-minus-up asymmetry, standardized by predetermined volatility so the common scale cancels, is the object that carries the paper's powered falsification of the leverage-cycle prior: where the Section 5.4 interaction is null but too imprecise to bound, standardizing here sharpens the test into a decisive equivalence verdict.

(i) Raw symmetry. The paired asymmetry $\Delta_h = \beta_h^{\text{down}} - \beta_h^{\text{up}}$, estimated on the same block resamples so the difference is properly paired, is statistically indistinguishable from zero at every horizon and tail level (Table 5, Panel A): at the 1% level on impact, $\Delta_0 = 0.00053$ ($p = .110$), with unstable signs across levels and horizons.

(ii) The standardized residual. Standardizing returns by predetermined volatility before forming the asymmetry measures does not remove a scale bias. A symmetric scale effect, predetermined or shock-induced, cancels in any down-minus-up difference, which is why the raw paired contrast in (i) is already null. What standardizing does is reduce the scale noise that masks a small shape residual, raising power. At the moderate tail the residual is null ($p = .446$ at the 5% level). At the extreme tail a small positive residual becomes statistically detectable: $\beta = 0.00071$ ($p = .012$) on the 1% tail indicator, with a winsorized third-moment regression also significant (-0.029). This sharpening, not any new channel, is why the residual surfaces here but not in the lower-powered raw contrast or in the Section 6.2 placebo, which compared the raw, vol-dominated gap. Net of volatility, the extreme downside is therefore not literally perfectly symmetric. The question that matters is whether the detectable residual is large enough to matter.

(iii) The bound. It is not. Converting the bootstrap intervals into minimum detectable effects and equivalence tests (Lakens, 2017; Riesthuis, 2024) against the pre-specified smallest effect size of interest (Section 4.5), the exceedance asymmetry (down minus up) at the 1% level is bounded below the smallest effect size of interest under all three calibrations of the shock span, including the strictest (MDE at 80% power 0.00094, below the strictest SESOI of 0.0011); the standardized residual is statistically detectable but economically negligible, clearing two of the three calibrations and borderline under the strictest (Table 5, Panel C, and Figure 8).[5] The resulting statement is the paper's central quantitative claim: we can rule out a downside-specific tail asymmetry larger than roughly 0.00094 per unit of the log liquidation shock, evidence of absence at the economically relevant scale, not mere absence of evidence (Altman and Bland, 1995; Ioannidis et al., 2017). The moderate tails (5%, 10%) are underpowered for strict equivalence, so the strong bound lives at the 1% level. That is also where the residual lives, so it is the bound that matters. The scope is narrow by design: the asymmetry the introduction documents is a far-extreme property, the $\pm 5\%$ and $\pm 7\%$ counts of Section 3, near the 0.1% tail that an hourly panel cannot resolve, so the bound concerns one-percent-tail downside specificity rather than amplification of that far-extreme asymmetry. And the third-moment regression is excluded from the equivalence exercise on unit grounds (a third moment is not a probability gap).

[5] Read as a continuous function of the smallest effect size rather than at the three pre-registered calibrations alone, the verdict does not turn on the choice of span: the exceedance Δ is equivalent-to-negligible for every SESOI above 0.0011, at or below all three locked spans, so it clears even the strictest; and the residual for every SESOI above 0.0012, which sits just above the strictest span and hence above only it, leaving the strictest calibration as the single inconclusive reading already noted.

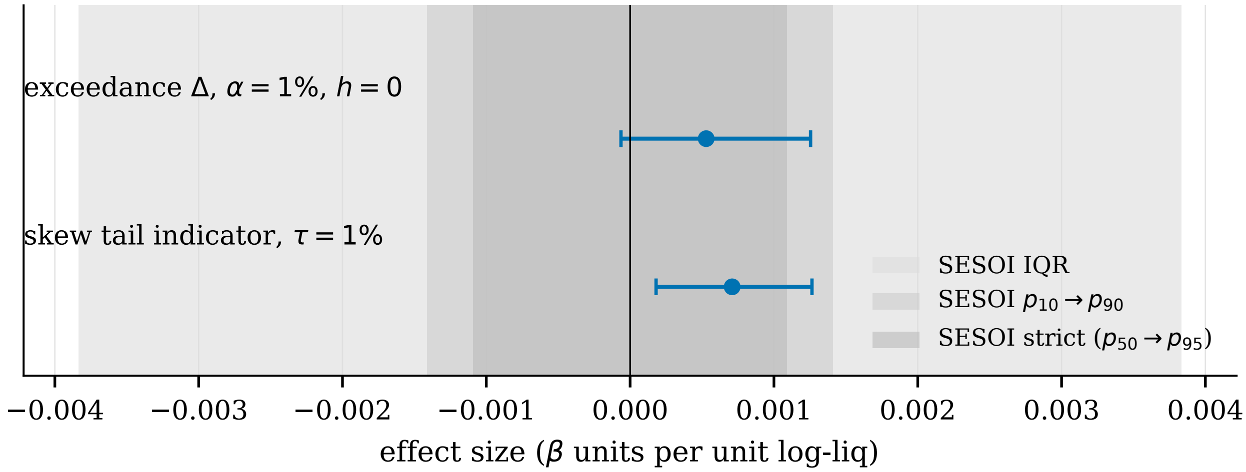

Figure 8: Point estimates and 95% intervals for the two 1%-tail asymmetry objects against the three SESOI bands. The residual is statistically detectable yet economically negligible.

The positive control. A bound is only as credible as the battery's ability to find what it claims to exclude, so we ran the full equivalence apparatus on a synthetic series into which a sign-specific downside cascade was planted at known scale. The headline object behaves as a calibrated instrument should: a planted asymmetry at half the smallest effect size of interest is returned as equivalent-to-negligible and goes undetected (point estimate 0.00026); at the threshold itself the verdict is inconclusive (0.00093); and at twice the threshold the asymmetry is recovered as non-negligible, its interval lying beyond the band (0.00244). Because the standard error is a within-panel block bootstrap anchored to the same 0.00094 rather than to the simulation budget, this graduation is a property of the data's information content, not of how long we ran the planting. The equivalence apparatus therefore detects a genuine, sign-specific downside asymmetry of roughly twice 0.0011 and is correctly silent below it. That is what converts the all-negative real result from "we found nothing" into the sharper "there is nothing at or above that scale to find." This certifies the power of the equivalence apparatus, the object that claims the bound; the symmetric placebo is deconstructive rather than a claim of absence and owes no such demonstration, though the same planting also escapes its band at and above that scale. One limit should be stated. The cascade is planted in the units the tests read, so the control certifies the battery's operating characteristics at a known, now economically anchored, effect size rather than re-detecting a skew that emerges from a structural process; a fully emergent cascade, a calibrated leverage spiral, is left for future work.

7.4 Stability

The surviving objects are not the artifact of one episode or one tuning choice. Splitting the sample in half leaves the tail profile essentially unchanged. Excising the August 2024 episode, the largest liquidation event in the sample, preserves about seventy percent of the twelve-hour deepening (-0.171 against -0.241) while the paired asymmetry stays null in every subsample (Table 6). The block-bootstrap inference, and hence the equivalence bound itself, is insensitive to the block length: re-running the critical cells at 12-, 36-, and 48-hour blocks moves the MDE by less than a tenth of its value (Appendix Table 9). One sample nuance is reported. The small positive impact tilt in Δ_0 is concentrated in the first half of the sample (2021–2023, the Terra/FTX era) and sits in the contemporaneous-reverse-causality zone, vanishing at $h \geq 1$ and in the later subsample.

7.5 What we do not claim

The guardrails. We do not claim that liquidations *cause* volatility or tail risk: at short horizons the relation is bidirectional, and every estimate is a predictive association. We do not claim the downside effect is literally zero: a 1% residual is detectable; we claim it is negligible and *bounded*. Symmetrically, the residual is not evidence of downside specificity: it is isolated (absent at the 5% level and in the raw exceedance contrast), below the smallest effect size of interest, and non-excluded rather than established. We do not claim OECD-style elasticities, only sign and order of magnitude. And we do not claim out-of-sample dominance over dedicated volatility models, only incremental information. What we do claim is the conjunction: a real, symmetric, bounded second-moment channel, validated in and out of sample.

8 Mechanisms

Sections 5–7 established an empirical fact: net of volatility, observable DeFi liquidations do not amplify ETH's downside tail to any economically significant degree. This section rationalizes that fact through market structure. The aim is not to re-estimate it, but to explain why an economist should not have expected otherwise at the observable DeFi scale. The argument is a beam, not a proof. No single channel demonstrates the null on its own, and the four are ordered by evidentiary weight, from the load-bearing channel (size) to the modifier (oracle). Each bounds one link in the hypothetical leverage-cycle chain, from liquidation to directional fire sale to permanent price impact to aggregate spiral, and the register stays descriptive throughout. The liquidation channel is real and locally disruptive.

It raises volatility, as Section 7.1 confirms; it is only too small and too efficiently cleared to aggregate into a directional downside loop.

8.1 Size of the liquidation flow

For a liquidation flow to imprint downside amplification on ETH's *aggregate* return distribution, it must be material relative to the turnover that forms that distribution. It is not. In the normal regime the median daily flow is on the order of 2×10^{-5} of one percent of ETH spot-plus-perpetual turnover (centralized-exchange reported), a rounding error; the mean is 0.0032%; and even the worst day of the entire sample reaches only 0.67% (Table 7). Against the largest reported single-day centralized-exchange liquidation cascade on record, roughly nineteen billion dollars on 10 October 2025 (CoinGlass, 2026), the typical daily DeFi flow sits four to six orders of magnitude lower, and the whole-sample cumulative total amounts to roughly 13% of that single cascade day. A flow this many orders of magnitude below the venue where a reflexive spiral plausibly lives (Section 9) is unlikely, at the observable scale, to be a material aggregate contributor to ETH's crash quantile. This size argument is conditional on observed coverage: the MakerDAO and liquid-restaking collateral that Section 2.7 documents as outside the measure do not enter the numerator, and the omission is bounded there as favorable to the null. In aggregate-market terms the hourly shock the quantile projection sees is negligible, and that is what predicts the null.

The inequality is wide, three to four orders of magnitude, wide enough to survive deflating the wash-inflated denominators entirely (the numerator is our own hard extraction; the caveat is detailed at the table). Because the numerator is debt forcibly repaid rather than collateral seized and sold, it excludes the liquidator's discount and so understates the volume that actually reaches the market, which makes the ratio a conservative bound on aggregate materiality. The flow is also concentrated in a single episode (August 2024). This is why the median, not the mean, carries the argument, and why we compare the typical flow to a cascade rather than the outlier day to a cascade. Agent-based work on DeFi lending finds systemic default rates below a tenth of a percent even at ten times observed volatility (Chaudhary and Pinna, 2022), and the ecosystem's lending base is itself non-monotone, peaking near $45 billion in late 2021 and falling to roughly $15 billion by mid-2023 (Bertomeu et al., 2024). "Small" is not only our empirical finding but what a model of on-chain lending predicts under stress.

8.2 Clearing

Even a small flow could imprint directionality if it discharged as a fire sale, with inventory sold under pressure, in size, into a falling market. The architecture of DeFi liquidation rules this out for the modal case. Liquidations are absorbed by well-incentivized liquidators (Qin et al., 2021); over seventy percent of liquidable positions are liquidated immediately, and eighty-five percent of liquidable collateral clears within two blocks (Perez et al., 2021), leaving no overhang to drain gradually into the price. And the modal liquidation is not a crash but a drift in position health. Moallemi and Patange (2024) document that over seventy percent of liquidations occur in the absence of any downward price jump. The central mode of DeFi liquidation is non-directional; the directional episodes are the tail, not the body. Toxic liquidation spirals do exist as a mechanism (Warmuz et al., 2022), but they are a conditional tail phenomenon (illiquid collateral, deep undercollateralization), not the modal case, and they do not contradict the rapid clearing of the typical liquidation.

8.3 Absorption and price impact

The next link requires the selling pressure to move the market price and stay there. Automated-market-maker depth and arbitrage absorb and reverse it instead. The measured price impact of a DeFi liquidation is minuscule, on the order of 0.05% on Aave and 0.01% on Maker at the block level, and temporary (Tian and Zhu, 2025). On the largest AMM, Lehar and Parlour (2025) document the absence of long-lived arbitrage opportunities, so a pool-level displacement is corrected rather than propagated into a persistent move in aggregate ETH. The most serious counterweight is the permanent price impact documented by Lehar and Parlour (2022), with spillover as far as centralized venues. But their own evidence bounds its reach for our asset, in three steps. The flagship contagion in their data runs through an illiquid collateral token, not through deep ETH; the arbitrage they document reverses the transient component; and their heavier tails are explicitly two-sided, with no quantile or asymmetry test. That is a level-impact result, not an aggregate downside-amplification one. For deep, hourly ETH, the aggregate footprint is dominated by the symmetric volatility channel of Sections 6–7, not by a directional translation of the distribution.

8.4 Oracle decoupling

The final link would require a transient on-chain price move to feed back into the aggregate market through the liquidation system itself. DeFi oracles are largely centralized-exchange-anchored and smoothed: they aggregate from external venues and are, in the words of the OECD study, oblivious to small price changes (Sasi-Brodesky and Nassr, 2023). A transient DEX deviation does not mechanically trigger a fresh liquidation wave priced off that deviation, so there is no small-move price-oracle-liquidation feedback loop. We place this channel last because its sign is design-conditional. The same oracle linkage is, in other configurations, itself a contagion channel, the broader oracle-design problem of Caldarelli (2020). For small transient moves on a deep asset, smoothing and anchoring decouple and attenuate; the oracle-as-cascade channel manifests on large moves of illiquid assets, consistent with the illiquid-token flagship of Section 8.3. The net effect is in our favor for ETH at the hourly grain, with the acknowledgment that the reverse mechanism exists, which is why this is a modifier and not a load-bearing channel.

The adversarial limit of this design-conditional channel is a liquidator who does not passively receive the oracle price but actively moves it. Sadeghi and Feinstein (2026) study profitable oracle manipulation that triggers liquidations on constant-product markets, a form of oracle-extractable value, and identify the transaction-fee conditions under which it becomes economically unviable. This is the precise micro-mechanism through which a liquidation-to-price loop could operate on chain, and a theoretical force that already bounds it. Our result bounds it from the data side: the hourly equivalence bound of Section 7.3 limits the asymmetric imprint such strategies, were they prevalent on deep ETH, leave in the aggregate tail, while staying silent on their block-level profitability, which an hourly panel cannot resolve. The active-liquidator premise is also ours; the flow we measure is cleared by agents who choose when to act.

8.5 Synthesis

The channel is real and raises volatility (Section 7.1), but too small relative to aggregate ETH turnover (Section 8.1) and too efficiently cleared (Sections 8.2–8.4) to aggregate into a directional downside loop in the observable on-chain channel. The null of Section 7 does not rest on this beam; it is established empirically and bounded by equivalence. But the beam makes it intelligible rather than surprising, in order of evidentiary weight: size (load-bearing, a hard number), then clearing, then absorption, then the design-conditional oracle modifier. The account is size-contingent. It explains why the channel does not aggregate *today*, not that it never could, a conditionality Section 9 takes up.

9 Scope, Mediation, and Limitations

9.1 Perimeter

What we tested is the observable on-chain channel. A reflexive leverage-cycle spiral, in the strict sense, plausibly lives on the centralized perpetual-futures venue, a compartment we do not observe and therefore neither test nor refute. Our null is not a denial of centralized-exchange cascades. It is a result about a different object, three to four orders of magnitude smaller. The scale gap carries this reading. The typical daily DeFi flow is four to six orders of magnitude below the largest single-day centralized cascade on record (roughly nineteen billion dollars, October 2025; Section 8.1), and the largest crashes of the period (Terra, FTX, October 2025) are, in event-study and practitioner accounts, associated with centralized, perpetual, or systemic dynamics rather than with on-chain DeFi. We are deliberate about the evidentiary status of that attribution. The weight of evidence favors the centralized venue, but the causal claim is associative on both sides: event studies, connectedness measures, and practitioner magnitudes, the very class of non-identified evidence this paper otherwise declines. The formulation we prefer is that the centralized venue is the most plausible locus, not an established cause, and that our DeFi design cannot adjudicate it. The supporting magnitudes are labeled as event-study and practitioner figures throughout, and the measured on-chain price impact remains minuscule and temporary (Tian and Zhu, 2025), real locally but too small to aggregate.

9.2 The null in the crypto leverage-effect debate

The field is split. In spot data at lower frequencies the leverage effect is inverted or absent (Baur and Dimpfl, 2018; Aharon et al., 2023), while the cascade narrative, in which liquidations drive the downside, lives in a different venue, frequency, and jump regime. Our null is the tail-response counterpart of the inverted-or-absent spot effect, transposed to a new object: net of volatility, the DeFi shock predicts large up-moves and large down-moves equally. It is not in tension with the cascade results, which inhabit an intraday, jump-driven, centralized regime that an hourly on-chain shock does not capture. The reconciling axis is the decomposition of Huang et al. (2022), in which the diffusive component behaves conventionally while jumps invert, so the net sign flips with the mix. That is why spot and cascade findings coexist without contradiction. Our contribution to the debate is one empirical point that sits in the both-tails-respond-more-than-the-mean camp (Ando et al., 2022). We do not adjudicate the true sign of the crypto leverage effect; we place our null within it. One apparent counter deserves a direct word. Cross-coin tail dependence is, if anything, stronger on the *upside* (Nguyen et al., 2020), but that is co-dependence across coins, not a single asset's response to a shock, and it is consistent with our finding of no downside specificity.

9.3 Scope of the null

The null must be stated precisely. At the observable on-chain scale and over this period, there is no economically significant downside amplification. That is not the same as saying there is none, or that there will be none. The conditional is explicit: *if* the DeFi share of crypto leverage were to grow materially, the channel *could* become aggregate-relevant. The weight of that statement is carried by three legs, size (Section 8.1), mechanism (Sections 8.2–8.4), and the official-sector consensus that DeFi is not yet systemic but bears watching as it grows (Financial Stability Board, 2023; Aramonte et al., 2021; International Monetary Fund, 2021; Hermans et al., 2022), never by the whisper. The conditionality is not rhetorical: on-chain leverage (Heimbach and Huang, 2024) and cross-protocol interconnection (Badev and Watsky, 2023) are each time-varying, and the ecosystem's measured fragility moves with them (Bertomeu et al., 2024). This is why we bound the null rather than assert a zero. The equivalence exercise (Section 7.3) separates two objects at the one-percent tail. Net of volatility, the exceedance asymmetry (down minus up) is bounded below the smallest effect size of interest under all three calibrations of the shock span, including the strictest, so it excludes any downside asymmetry larger than roughly 0.00094 per unit of shock; a small standardized residual is statistically detectable but economically negligible, clearing two of the three calibrations and borderline under the strictest. Both readings make the null respectable and falsifiable rather than a bare failure to reject (Altman and Bland, 1995; Ioannidis et al., 2017).

The one-percent whisper bounded in Section 7.3 is, in addition, governed at $\tau = 0.01$ by a non-Gaussian limiting distribution that makes standard inference unreliable to begin with (Chernozhukov, 2005; Chernozhukov and Fernandez-Val, 2011). That is one more reason it does not carry the conclusion, which rests on size, mechanism, and consensus.

9.4 Limitations

We list the limitations, say why each is one, and state what was done about it. *(1) Hourly aggregation.* The shock is aggregated to the hour, so within-hour sequencing is lost and the impact hour mixes trigger and consequence; we adopt a predictive register and read across horizons, but no tick-level sequencing is available. *(2) BTC–ETH co-movement.* The impact response replicates on Bitcoin (Section 6.6: $\beta_0^{\mathrm{BTC}}(0.01) = -0.023$ against -0.032 for ETH), so the shock is partly a common-crypto stress proxy rather than purely ETH-specific; the cross-asset placebo is precisely the diagnostic for this, and the result is what motivates retiring the specificity claim rather than defending it. *(3) Short-horizon reverse causality.* 85.5% of the worst-1% hours contain contemporaneous liquidations against 41.6% of the best, so a crash mechanically triggers liquidations and the short-horizon relation is bidirectional. Hence the non-causal register, with the one-hour lag and horizon reading as mitigation. *(4) No centralized exchange.* The perimeter is on-chain only, so the plausible locus of the spiral is structurally out of reach. *(5) Mean, not quantile, residualization.* The orthogonalized shock is the residual of an OLS, not a quantile, regression. We acknowledge this as a refinement, with the raw in-regression controls serving as the canonical channel and the orthogonalized shock reserved for the placebo. *(6) Estimated denominators.* The size ratio rests on centralized-reported, partly wash-inflated turnover figures, so it is an order-of-magnitude bound; the numerator is the only hard figure, and the inequality survives one to two orders of rescaling. *(7) No structural identification.* Every estimate is a predictive association; converting them to causal statements would require exogenous variation in liquidation timing, the most immediate direction for future work. Limitations (1)–(3) bound causality, (4) bounds scope, and (5)–(7) bound precision; none invalidates the beam, and together they fix the perimeter of the claim. The coverage limitations of the underlying data (MakerDAO de facto absent, the excluded liquid-restaking collateral, Bybit-only open interest, the oracle-versus-perpetual price mismatch, and the dust and exploit filters) are established and bounded in Section 2.7, and each was shown there to leave the null intact or to sharpen it.

10 Conclusion

This paper began with a real puzzle and a plausible prior. Ethereum's hourly returns carry a marked downside tail asymmetry, the textbook leverage-effect explanation is absent or inverted in crypto, and the field falls back on the liquidation cascade of the leverage cycle, a mechanism that DeFi, alone among markets, records transaction by transaction. A naive quantile local projection appears to confirm the prior in full: a first-percentile response 18.8 times the median on impact, deepening with the horizon, robust across a battery of robustness checks, surviving multiple-testing correction.

A battery of diagnostics overturns that reading without overturning the estimate. The coefficient is genuine in level. Both permutation nulls leave mechanical overlap a minority share, with no signed bias. But its apparent downside specificity does not survive. A symmetric, zero-skew placebo reproduces the entire left–right tail gap at every horizon; the per-period construction shows both tails loading equally; the headline ratio is an artifact of an immobile median; and the whole signature replicates when Bitcoin is substituted as the outcome. What the naive analysis detected is a real but symmetric volatility channel, generic to crypto stress, dressed up as downside fragility. This is the trap that symmetric heteroskedasticity sets for quantile projections on overlapping cumulative returns, here sprung on real data where the naive answer looked spectacular.

What survives is modest and worth having. Liquidations are a real, symmetric predictor of ETH tail risk, in and out of sample. The effect is incremental to a nested quantile benchmark, not a replacement for a dedicated scale model; the volatility channel they drive confirms and extends the existing transaction-level evidence to the conditional tails; and, net of volatility, the one-percent-tail exceedance asymmetry (down minus up) is bounded below the smallest effect size of interest under all three calibrations of the shock span, including the strictest, so we can *exclude* downside-specific amplification larger than roughly 0.00094 per unit of the log-liquidation shock. That is evidence of absence at the economically relevant scale for the one-percent-tail exceedance-asymmetry object under the stated calibrations of the shock span, not a bare failure to reject. The small standardized residual at the one-percent tail is statistically detectable but economically negligible, clearing two of the three calibrations and borderline under the strictest, and it does not carry the conclusion.

The takeaway is bounded on every side. At the observable on-chain scale and over this period, DeFi liquidations do not amplify ETH's downside tail beyond their symmetric volatility effect; they nonetheless predict tail risk, symmetrically; and a naive quantile local projection overreads that symmetric volatility as directional fragility. This is not no effect: the volatility channel is real. The short-horizon relation is bidirectional, so the finding is not causal. It does not refute centralized-exchange cascades, which our design cannot observe. And it does not claim that ETH's asymmetry is fictional. That asymmetry is real; the observable DeFi channel simply adds no downside-specific amplification to it, net of volatility. The null is size-contingent: if the DeFi share of crypto leverage grows materially, the channel could become aggregate-relevant, and the bound bears re-examination as that share moves.

Four directions follow. We ran an unconditional-quantile (RIF) cross-check (Section 6); because RIF-OLS is here algebraically tied to the conditional model, a fully *independent* unconditional estimator would test the artifact from a separate angle. An identification strategy isolating exogenous variation in liquidation timing, whether an adjustable-rate instrument or residualization at the quantile rather than the mean, would license the causal statements this paper declines. And centralized perpetual-futures data would let one test the spiral where it plausibly lives, outside our perimeter. And broadening the on-chain liquidation panel beyond the currently indexed lending protocols, to MakerDAO and DAI-vault liquidations and to liquid-restaking collateral (Appendix C), would close the two coverage gaps whose direction we can bound but not eliminate here. The contribution we claim is correspondingly sized: a reproducible on-chain liquidation panel, a portable methodological caution that symmetric volatility masquerades as fragility in naive quantile projections, a bounded null against a popular directional prediction, and a symmetric tail-risk predictor, executed and stress-tested in full.

Declaration on the use of generative AI

This project made substantial use of generative artificial intelligence: large language models of the Anthropic Claude family (the Opus and Fable generations, 2026), operated through an agentic coding interface with multi-agent orchestration, under my direction and subject to my review at every stage. Their role spanned several parts of the work: discovering and organising the related literature and assembling the bibliography; proposing diagnostic and robustness strategies, analytical framings, and candidate interpretations; scaffolding and drafting analysis and packaging code; executing the pre-committed estimation pipeline to regenerate the deposited artifacts; building the verification and reproducibility infrastructure of the replication package; and drafting and revising targeted passages of the manuscript. Throughout, I treated these tools as fallible research aids rather than as sources of authority, and subjected their output to a verification protocol commensurate with their documented failure modes.

The integrity of the results does not rest on trust in that assistance. The research question, the locked specification, the identification and inference strategy, the data sources and their construction, and every modelling and interpretive decision are my own; the specification was locked before the diagnostic phase. No reported quantity was estimated by a language model. All estimates derive from my seeded, deterministic pipeline, which the assistant executed only to regenerate the committed artifacts; the regenerated output was verified identical to my originals at reporting precision, and every in-text numerical macro is byte-matched by a non-regression guard. Two points warrant note. First, the size shares in Table 7 reflect a change of denominator basis that I decided and the assistant implemented: deterministic arithmetic over author-verified industry constants, not estimation, isolated and verified to move exactly those quantities, with the prior convention retained as an alternative row. Second, quantities quoted from primary sources were inserted only after verbatim checking against those sources. Correctness is therefore established by reproduction rather than by provenance: a single command re-derives every table and figure from the deposited data on a fresh clone, the guard confirms that all reported numbers are unchanged after any code change, and AI-assisted modifications were accepted only when the guard, the test suite, and a byte-level comparison of independently regenerated artifacts agreed that the scientific content was unchanged.

The characteristic failure mode of language-model assistance is text that is fluent but unsupported, and it was treated accordingly. The verification layer identified and removed several fabricated or mis-attributed citations, reworded or cut statements that could not be traced to a primary source or to a re-executable computation, and caught coding errors before they reached the results; I report this as evidence that verification was both necessary and applied. The build automation, verification apparatus, and data-acquisition scripts of the replication package are likewise substantially AI-authored under my direction and approved by me, and the committed result files were regenerated under a pinned deterministic environment. These models do not meet the criteria for authorship. I reviewed and approved all content and assume sole responsibility for this paper, including any remaining errors.

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Table 2: Diagnostic tests, one per confound.

Diagnostic	Confound isolated	Logic
Symmetric sign-flip placebo	volatility (scale)	does a zero-skew world with the real volatility path reproduce $\beta(0.01)$ and the left–right gap? (Wu, 1986; Liu, 1988; Davidson and Flachaire, 2008; Carriero et al., 2024)
Dual permutation nulls	overlap artifact	destroy the shock–return link (circular shift; innovation reshuffle), keep the cumulative LHS: any surviving β is mechanical (Hodrick, 1992; Boudoukh et al., 2008; Valkanov, 2003)
Tail-exceedance regressions	(the predictor + the symmetry test)	does the shock predict future tail violations, and equally on both sides? (Engle and Manganelli, 2004; Balcilar et al., 2017)
Standardized skew tests	scale noise	standardize by predetermined volatility, then test down-minus-up and the third moment (Neuberger and Payne, 2021; Farago and Hjalmarsson, 2023)
BTC-outcome placebo	asset specificity	re-run the full QLP with BTC as the outcome: does the signature replicate on an asset without the ETH-collateral channel?
Equivalence bounds (MDE/TOST)	power	convert "not found" into "we exclude effects larger than X" (Altman and Bland, 1995; Ioannidis et al., 2017; Lakens, 2017; Riesthuis, 2024)
Out-of-sample pinball test	in-sample overfit	does liquidation information improve rolling-origin tail forecasts against nested and GARCH benchmarks? (Diebold and Mariano, 1995)
Planted-cascade positive control	power (false negatives)	does the battery recover a sign-specific downside asymmetry planted at known scale, and stay silent below it?

Table 3: Quantile local projections: $\beta_h(\tau)$ of cumulative ETH returns on the liquidation shock.

τ	$h=0$	$h=1$	$h=3$	$h=6$	$h=12$	$h=24$
0.01	-0.032	-0.087	-0.105	-0.160	-0.241	-0.213
0.05	-0.019	-0.045	-0.046	-0.079	-0.102	-0.116
0.10	-0.013	-0.030	-0.034	-0.043	-0.061	-0.095
0.50	0.002	0.006	0.010	0.014	0.032	0.038
0.90	0.014	0.028	0.038	0.047	0.070	0.102
0.95	0.018	0.031	0.044	0.041	0.077	0.116

Notes: Locked specification (seven regressors, Section 4); $N \approx 40\,582$. Point estimates; tail inference is by moving-block bootstrap (Figure 3 displays the 95% band on $\tau = 0.01$), kernel standard errors being unreliable at extreme quantiles (Appendix Table 8).

Table 4: Out-of-sample pinball skill of liquidation information (%), against a nested quantile benchmark (top panel) and a dedicated GARCH(1,1) scale model (bottom panel).

	$h=1$	$h=3$	$h=6$	$h=12$	$h=24$
vs. nested QR (no liquidation terms)					
$\tau = 0.01$	1.41	1.03	1.10	1.18*	0.81
$\tau = 0.05$	0.97***	0.46**	0.36*	0.47**	0.15
$\tau = 0.95$	0.85***	0.56**	0.37*	0.26	0.60***
$\tau = 0.99$	0.51	0.54	1.11**	0.03	0.03
vs. GARCH(1,1) benchmark					
$\tau = 0.01$	0.33	-0.78	0.51	2.14*	1.59
$\tau = 0.05$	-0.61	-0.61	-0.16	-0.19	1.00
$\tau = 0.95$	-1.47**	-1.07	-0.59	-0.50	1.42**
$\tau = 0.99$	-0.44	0.41	1.38	0.29	2.51*

Notes: Rolling-origin forecasts of the per-period τ-quantile of ETH returns; expanding window, weekly refits, 97 re-estimations, ~16,200 test hours. Skill = $1 - L_{\text{full}}/L_{\text{bench}}$ in percent; stars from Diebold–Mariano tests with HAC variance (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). Source: `oos_predictive.csv`.

Table 5: Paired asymmetry, standardized skew, and equivalence verdicts.

Panel A: paired exceedance asymmetry $\Delta_h = \beta_h^{down} - \beta_h^{up}$, $\alpha = 1\%$ (per-period)				
h	Δ	95% CI		p
0	0.00053	[-0.00006, 0.00125]		0.110
1	0.00029	[-0.00045, 0.00097]		0.428
3	-0.00023	[-0.00088, 0.00033]		0.466
6	0.00015	[-0.00062, 0.00090]		0.694
12	-0.00007	[-0.00087, 0.00072]		0.844
24	-0.00044	[-0.00112, 0.00023]		0.234
Panel B: vol-standardized skew tests (h=0)				
measure	β	95% CI		p
tail indicator, τ=5%	0.00043	[-0.00075, 0.00158]		0.446
tail indicator, τ=1%	0.00071	[0.00018, 0.00126]		0.012
winsorised z^3	-0.02930	[-0.04391, -0.01479]		<0.001
Panel C: equivalence (TOST) verdicts vs. SESOI spans				
object	MDE_{80}	IQR span	strict span	$p_{10} \to p_{90}$
exceedance_delta (α=0.1 (h=0))	0.00231	equiv.	inconcl.	inconcl.
exceedance_delta (α=0.05 (h=0))	0.00171	equiv.	inconcl.	equiv.
exceedance_delta (α=0.01 (h=0))	0.00094	equiv.	equiv.	equiv.
skew_test (skew_tail05)	0.00166	equiv.	inconcl.	equiv.
skew_test (skew_tail01)	0.00077	equiv.	inconcl.	equiv.
skew_test (z3_winsor)	0.02081	n/a	n/a	n/a

Notes: Panel A: paired moving-block bootstrap (1,000 replications, 24-hour blocks). Panel B: returns standardized by predetermined 7-day realized volatility. Panel C: MDE at 80% power and TOST equivalence verdicts against the three SESOI calibrations of Section 4.5. Sources: `exceedance_paired.csv`, `skew_test.csv`, `mde_equivalence.csv`.

Table 6: Subsample stability: the tail profile and the paired asymmetry.

	$\beta_h(0.01)$				Δ at h=12	
Subsample	$h=0$	$h=6$	$h=12$	$h=24$	estimate	MDE_{80}
Full sample	-0.032	-0.160	-0.241	-0.213	-0.00007	0.00112
First half	-0.030	-0.175	-0.229	-0.214	-0.00005	0.00157
Second half	-0.037	-0.174	-0.292	-0.096	0.00017	0.00148
Excl. Aug-2024	-0.032	-0.127	-0.171	-0.165	-0.00002	0.00107

Notes: $\beta_h(0.01)$ point estimates per subsample; Δ at h=12 with its 80%-power MDE from the paired bootstrap. "Excl. Aug-2024" drops a window around 2024-08-05 extended backward by the maximal horizon, so the episode contaminates neither regressors nor outcomes. Source: `subsample_stability.csv`.

Table 7: Size of the DeFi liquidation flow relative to ETH turnover and to centralized-exchange cascades.

	Share (%)
Mean daily DeFi liq. / daily ETH turnover	0.0032
Median daily DeFi liq. / daily ETH turnover	0.0000
Worst day (2024-08-05) / daily ETH turnover	0.67
Worst 7 days / 7-day ETH turnover	0.10
Worst day / Oct-2025 CEX-perp cascade ($\sim$\$19bn)	1.62
Whole-sample total / Oct-2025 single-day cascade	13.3

Notes: Numerator is our measured flow (the hard number); denominators are industry turnover estimates (CoinGecko, 2025a,b) and a reported cascade magnitude (CoinGlass, 2026), partly inflated by wash trading (Gan et al., 2022) and not peer-reviewed, hence order-of-magnitude bounds. The inequality is wide enough to survive any plausible rescaling of the denominator. Source: `size_ratio.csv`.

A Robustness Battery

The conventional robustness battery referenced in Section 5 comprises a cross-asset placebo (the orthogonalized shock applied to XRP and DOGE returns, neither used as ETH-collateral), alternative shock definitions (raw versus BTC-orthogonalized), subperiod and crisis-episode exclusions, a monotonicity test across quantiles, and the exact-specification bootstrap band of Figure 3. The sign and tail significance of $\beta_h(0.01)$ and $\beta_h(0.05)$ survive throughout, while the leverage interaction is null in every variant. This is the pattern of Section 5.4. The full battery and its companion metadata are in the replication package (`robustness_*.csv`).

Dual permutation nulls. The two nulls of Section 6.3 are reported jointly, with their preservation diagnostics, in `pure_null_circular_shift_by_horizon.csv` and `pure_null_innov_shuffle_by_horizon.csv`. The conservative circular-shift null preserves the shock's marginal distribution and autocorrelation exactly (verified in the metadata) while severing its alignment with returns; the innovation-reshuffle null destroys the return's serial dependence (its absolute-return autocorrelation falls from its empirical value to essentially zero). Both yield a minority artifact share at every horizon (0.18 and 0.08 at twelve hours), with a signed null mean indistinguishable from zero (0.004 under the conservative null), and the real coefficient outside the null band throughout. Both constructions are small, and neither produces a signed downward bias.

Nine-quantile grid. The main text reports the six-quantile grid $\{0.01, 0.05, 0.10, 0.50, 0.90, 0.95\}$; the nine-quantile grid (adding 0.25, 0.75, 0.99) is provided in `quantile_lp_results_9q.csv` and yields the same qualitative profile, with the 0.99 upside mirror used for the left–right gap statistic of Section 6.2.

Table 8: Kernel versus bootstrap standard errors, same specification, $\tau = 0.01$.

h	kernel SE	bootstrap SE	ratio
0	0.0057	0.0094	1.65
3	0.0229	0.0362	1.58
6	0.0344	0.0931	2.71
12	0.0404	0.1346	3.33
24	0.0465	0.0930	2.00

Notes: Bootstrap standard error from the exact-specification bootstrap; kernel standard error from the main table. Source: `se_ratio_nb07.csv`.

Table 9: Block-length sensitivity of the equivalence bound (MDE at 80% power, Δ at $\alpha =$ 1%).

h	block = 12h	block = 24h	block = 36h	block = 48h
0	0.00096	0.00094	0.00089	0.00091
1	0.00101	0.00101	0.00096	0.00097
3	0.00087	0.00087	0.00087	0.00088
6	0.00113	0.00109	0.00097	0.00096
12	0.00108	0.00114	0.00112	0.00114
24	0.00102	0.00097	0.00099	0.00101

Notes: Per-period paired bootstrap, 1,000 replications. The 24-hour column reproduces the main exceedance results. Source: `block_sensitivity.csv`.

B Tail Inference

Same-specification standard-error ratio. The case for bootstrap inference at the tail (Section 4.4) rests on a same-specification comparison. We take the bootstrap standard error from the exact headline-spec bootstrap and the kernel standard error from the same table, at $\tau = 0.01$ (Table 8). The ratio ranges from 1.6 to 3.3 across horizons. This compares like with like. A cross-specification comparison would instead conflate the standard-error gap with differences in scale, controls, and shock definition.

Block-length sensitivity. The 24-hour block length of the moving-block bootstrap is the maximal horizon, the natural choice for the overlap structure. Nothing in our results turns on it. Table 9 reports the 80%-power minimum detectable effect for the paired one-percent asymmetry, the object that drives the equivalence bound, at block lengths of 12, 24, 36, and 48 hours. The bound moves by less than a tenth of its value across the range. By construction the 24-hour column reproduces the corresponding cells of the main exceedance table exactly, a built-in cross-check that the two run on identical machinery.

C Construction of the Liquidation Measure

The liquidation series is extracted from Dune Analytics query `6912877`, versioned in the replication package, over the open-source Spellbook tables `lending.borrow` (debt leg) and `lending.supply` (collateral leg), across all EVM chains and lending protocols indexed at extraction time. This appendix records the construction decisions that materially affect the primary regressor. The full query is in the package.

Pre-aggregation before the join. An event-level join of the debt and collateral legs on transaction produces a Cartesian product for transactions containing multiple debt and multiple collateral events. This pattern is common in bundled liquidation bots, and it inflates totals. The query pre-aggregates each leg by transaction before joining, eliminating the inflation by construction. The correction revised the 2024 and 2025 aggregate volumes downward by roughly fourteen and thirty percent respectively relative to a naive join, while pre-2024 aggregates moved by less than a quarter of a percent.

Primary measure. The primary variable is the hourly sum of debt forcibly repaid in USD, zero-filled on hours without qualifying liquidations, entered as $L_{t-1} = \ln(1 + \text{liq}_{t-1}^{\text{USD}})$ with a one-hour lag. The seized-collateral leg is retained as a robustness alternative. The two measures correlate above 0.97, and the choice changes the headline coefficient by less than three percent in log space.

Collateral filter. Only liquidations seizing ETH-like collateral are kept, using the symbol set {WETH, ETH, stETH, wstETH, rETH, cbETH, sfrxETH, ETHx}, so the measure isolates the ETH leverage channel. Liquid-staking derivatives are included, so the collateral is ETH-like in the broad sense. A depeg of a staking derivative could in principle inject non-pure-ETH liquidations, a rare event we note for completeness.

Exclusions. Three exclusions are applied on top of the structural filters. Positions with absolute debt below five dollars are dropped, removing a cluster of roughly 75,000 micro-events in September 2025 linked to bot activity that is immaterial in dollar terms. UwuLend liquidations from June 2024 onward are dropped to exclude an exploit-driven hour reporting an implausible volume attributable to on-chain exploit movements combined with pricing errors on illiquid tokens. Euler liquidations during the March–April 2023 hack window are dropped; Euler v2, a distinct protocol launched in 2024, is retained. A small number of hours in late 2025 carry anomalous collateral-to-debt ratios from oracle pricing of illiquid tokens; these are economically invisible in the debt-repaid measure.

Coverage caveats. MakerDAO, historically the largest on-chain liquidator, sits outside the Spellbook lending schema (a CDP architecture) and is de facto absent, so the measure understates total on-chain deleveraging, most severely in 2021. No centralized-exchange liquidations enter. The query was executed once, on a single date; a future re-execution may differ marginally if the Spellbook tables are updated.

Sensitivity to the MakerDAO gap. Because the MakerDAO omission is the largest coverage gap, we bound its effect by an actual re-fit rather than an assumption. We inflate the observed liquidation series by a missing-Maker share m (up to a punitive $m = 0.5$) and re-estimate the locked specification, placing the missing mass three ways: proportional to currently-active hours, on stress-adjacent zero hours, and weighted toward high open-interest (leverage) hours. The shock coefficient's sign is preserved under every allocation and horizon. At impact its magnitude moves by at most 24%, but at twelve hours the leverage-weighted infill (the largest-displacement allocation, since Maker auctions cluster in high-leverage, falling-price hours) attenuates the deepening by up to 65%. Both displacement figures are the maximum over the defensible allocation range $m \leq 0.25$; the punitive $m = 0.5$ end of the sweep changes neither sign nor significance. The medium-horizon magnitude is therefore coverage-sensitive. The bounded object is not: the per-period exceedance asymmetry at the 1% level stays within 0.48 of the smallest effect size of interest under every allocation, and shrinks toward zero under the leverage-weighted and stress-adjacent infills, so a more complete measure would tighten rather than overturn the null.